The Importance of Inter Vivos Transfers: Insuring and Investing in Children

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Abstract

We argue that inter vivos transfers from parents to children are much more important than previously thought for the cross-sectional dispersion of income and wealth as well as the intergenerational persistence of those quantities. We document two common features of transfers, front-loading and insurance, that account for their power. We also show how the size and type of transfers made to children vary across the wealth distribution and develop micro-level estimates of the effects of different types of transfers on children's choices. We establish the importance of these features for macroeconomic outcomes by embedding them in a quantitative overlapping generations (OLG) model with incomplete markets, education choice, and human capital accumulation.

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1 Introduction

It's a popular and well-documented fact that parents play a large role in shaping their children's economic outcomes. We observe a great deal of similarity between parents and children in their decisions regarding consumption, saving, education, labor-supply, portfolio composition, and other economically-relevant variables,[1] a similarity that exists in spite of the massive cross-sectional dispersion apparent in such quantities. It is so suggestive of parental influence, which was already an intuitive and salient factor in the minds of both researchers and the general public, that a very large and very old literature formed to understand the ways in which parents affect their children's economic lives.

Among the many channels that have been studied, transfers receive less attention as a source of persistence in outcomes and, within that category, inter vivos transfers (IVTs) receive even less. There are reasonable explanations for why they have been somewhat neglected. For one, compared to bequests, they are smaller in magnitude. Average annual bequest payments stand at $476bn while parents disbursed $160bn in inter vivos transfers, excluding educational transfers.[2] For another, there is a substantial body of work[3] establishing the considerable power of early-childhood investments and parenting decisions to determine children's long-run outcomes, leading to a reduced focus on parental investments in adult children.

While bequests and the circumstances of one's upbringing are rightfully at the center of discussions on the variation in and persistence of consumption, income, and wealth, we wish to argue that inter vivos transfers in adulthood belong there as well.

Though smaller than bequests, they are still quite sizable in the aggregate: if they were to become a separate budget function, inter vivos transfers would represent one of the ten largest paid for by the US Federal Government.[4] Moreover, empirical work by Fagereng et al. (2021) using quasi-random variation in assignment of adoptees in Norway already documents a significant role for inter vivos transfers in the persistence of wealth levels across generations.

[1] Foremost is Charles and Hurst (2003). A selection of other papers documenting intergenerational persistence: Couch and Dunn (1997), Belzil and Hansen (2003), Restuccia and Urrutia (2004), Waldkirch et al. (2004), Charles et al. (2014), Cronqvist and Siegel (2015), Boserup et al. (2016), Fagereng et al. (2020), Coda Moscarola et al. (2025).

[2] According to our calculations using Health and Retirement Survey data from 1996 to 2019, in which we observe the transfers made by parents over the age of 50. These numbers are somewhat larger than those documented in Feiveson and Sabelhaus (2018) using Survey of Consumer Finances (SCF) data.

[3] Surveyed in Heckman and Mosso (2014) and Doepke et al. (2019).

[4] It would rank eighth, slightly higher than transportation, which stands at $145bn in spending per year (PGPF 2026) and, among the income security programs, it would be significantly larger than unemployment insurance (UI) and slightly larger than Food and Nutrition Services (CBO 2024).

We build on this body of work first by documenting features of inter vivos transfers across the wealth distribution. We argue that they have several characteristics that generate large indirect effects, which help them punch above their already substantial weight. We document that they are *front-loaded*, occurring at the beginning of children's careers and *provide insurance*, paying out primarily in response to unexpected increases in expenses or decreases in incomes. Second, we provide micro-level estimates of the effects of transfers on children's decisions. Third, we embed these channels in a structural overlapping generations (OLG) model to understand their impact on aggregate variables and intergenerational dynamics.

Front-loading of transfers. Front-loading transfers increases their value to the children who receive them. The simplest reason is that, in present-value terms, a transfer received earlier is worth more than a transfer of the same nominal value received later. Beyond that, transfers that occur earlier also occur in periods where their marginal value is highest, which magnifies their impact on saving and investment decisions. Both have effects that are felt for years in the future, so earlier investments have benefits that accrue over longer periods of time, which is another feature that accounts for the power of front-loading.

We assess the degree of front-loading by computing the paths of cumulative transfers received by children over their adulthood. Certainly, educational transfers come sooner than any others, but we show that, even excluding educational transfers, a majority of the transfers children receive come before their parents reach retirement age. Moreover, the vast majority of all transfers are given within the first ten years of children's adult lives, when their effects on children are the largest.

We also illustrate how earlier receipts of inter vivos transfers magnify their importance relative to bequests by computing present values of transfers and bequests for different wealth groups, from the perspective of an eighteen year-old. In all cases with a discount rate of 2%, inter vivos transfers are comparable in value to bequests and represent the majority of transfer wealth[5] for children of parents in the bottom 75% of the wealth distribution. When increasing the discount rate to 5%, as the literature argues that rates of return are higher for the wealthy,[6] inter vivos transfers represent the majority of transfer wealth for all wealth groups. Further increases in the discount rate only make the comparison more extreme, with early transfers and early income accounting for even larger

[5]Transfer wealth being the sum of present values of bequests and inter vivos transfers.

[6]See Fagereng et al. (2020) for evidence from Norway and Bach et al. (2020) for evidence from Sweden. Kuhn et al. (2020) also demonstrate the importance of high returns obtained wealthy individuals for explaining trends in wealth inequality in the US. See also Saez and Zucman (2016), Smith et al. (2023), Bricker et al. (2025), and Bhandari et al. (2026).

shares of total wealth.

Insurance via transfers. The insurance function of intergenerational transfers also provides a boost to the effectiveness of inter vivos transfers. From the child's perspective, these transfers are valued more highly than their nominal amount would suggest because they receive them in states of the world where their marginal value of wealth is high. Then, because they're concentrated in states where children are constrained or very-nearly constrained, insurance transfers induce greater behavioral modifications than transfers in higher income states.

Yet, even if parents never actually make any insurance transfers, the whole risk/return calculus of children's human capital investment is changed by virtue of the fact that they would have been paid out if needed. Eliminating some negative states makes the same risky investment more attractive for children *ex ante*. Additionally, because parents' insurance substitutes for children's self-insurance, it also changes the balance between saving and investing. Partially freed up from the need to accumulate buffer stock savings early in their careers, children can target steeper income paths, ones that start with lower incomes but are ultimately more lucrative.

To study the degree of insurance provided by parents to children via transfers, we look at the transfers given in response to adverse changes in children's incomes and expenditures. In particular, we want to know when the changes are large enough to trigger payments and how much of the adverse change is absorbed by the corresponding transfers. We also want to know how to value these transfers from the perspective of the child.

We find that the changes in net income required to prompt a transfer are fairly large, at least on the order of going from full-time to part-time work, though small enough that they're not uncommonly observed in the data. Though, on the other hand, the states in which parents step in to help children vary across the wealth distribution, with wealthier parents providing assistance sooner than poorer parents, consistent with the fact that they have the resources with which to do so.

Then, when transfers are ultimately made in response to these events, we find that the replacement rate is significant. For parents in the top quartile of the wealth distribution, the average insurance transfers made to children ages 18-30 after these events ranged from \$2,265 to \$4,652. Of course, these estimates obscure a large degree of heterogeneity in parents' generosity within wealth groups. Among parents responsive to children's income changes, the number is even larger.[7]

[7]Compared to the estimates in Andersen et al. (2020), which are based on highly detailed transaction-level data from Denmark, ours are much larger. However, given that Denmark's social safety net is much

The next part of our analysis is aimed at understanding the effects of parental insurance on the decisions made by children. In particular, we focus on how the amount of risk children take on is impacted by the provision of parental insurance, which takes the form of saving on behalf of children, named *dynastic precautionary saving* in Boar (2021).

Because dynastic precautionary saving responds to children's risk levels even as it affects them, we instrument for it using parents' wealth levels before their children reach adulthood. Following this approach, we find that parental insurance leads to higher levels of equity ownership, business ownership, and permanent income risk. That is, parents' decisions to insure their children incentivize their children to take on a larger degree of productive risk. Consistent also with the hypothesis of parental insurance partially substituting for children's self-insurance, we find that children's saving rates are decreasing in the amount of savings held by their parents.

Propagation of transfer heterogeneity. The amplification of the effects of transfer payments that comes from front-loading and insurance also, therefore, amplifies heterogeneity in parents' transfer policies. This is especially striking because we already observe considerable differences in transfers both within and across wealth groups, differences that would be non-negligible even without amplification.

While parents generally prefer, regardless of their own individual characteristics, to make transfers in their children's early adulthood, the amount of transfers they make increase strongly with parents financial resources. We also find that, while the overall mix of transfers is very similar across wealth groups, insurance transfers account for a larger share of transfers made by parents in the highest wealth group.

Our approach to understanding the importance of these channels for cross-sectional variation and intergenerational persistence of income, wealth, and consumption is structural. We start by specifying a life-cycle model with human capital accumulation, endogenous college attendance decisions, incomplete markets, and idiosyncratic income risk. We add to this a Markov process for the parameters of the human capital accumulation function, including the starting level of human capital, in order to capture the factors of intergenerational persistence that come from investments in children that are made before they enter adulthood, as well as any hereditary factors.

The model's defining feature, which makes it suitable for our purposes, is that we allow for variations in transfer functions for each agent in the model: cross-sectional differences in the size and type of transfers each person can expect to give and receive over

more generous than its US equivalent, it makes sense that informal insurance would play a larger role in the US.

their life spans. We allow for the distribution of transfer functions to depend on the parameters of a child's human capital process, so that the model, if it is necessary to fit the data, can generate a degree of persistence in transfers across generations. We also allow the distributions of transfers received to depend on the children's college attendance to capture the possibility that parents may try to incentivize their children to attend college by committing to more generous transfer functions, given attendance.

We choose the parameters of the Markov processes in order to match the average income and transfers received by the children of each wealth bucket over their lives. We also fit the share of total transfers accounted for by each type. Then having a model that can match the relevant conditional moments, we're able to assess the importance of transfers in explaining the persistence of wealth and income ranks.

In effect, we can decompose the persistence and variability into portions determined by other factors, like early-life investments, and portions determined by transfers. We may also change the distribution of transfers over time, ask how reductions in front-loading or reductions in insurance would change the effectiveness of transfers for shaping children's behavior. More concretely, we can understand how policies that incentivize a particular distribution of transfers over children's lifetimes, such as stepped-up basis, also shape children's outcomes.

The results of our exercises suggest that transfers are highly consequential and that, indeed, the timing of transfers is essential for their power. We find that income levels would be uniformly lower without inter vivos transfers by around $20,000 in present-value terms without general equilibrium price responses and $15,000 with them. Moreover, eliminating transfers would increase wealth mobility among for all but the bottom quartile, for whom self-financing of human capital accumulation is more difficult. Consistent with this, we also find that further front-loading would increase persistence, but primarily at the top of the wealth distribution.

Related literature. Our paper fits broadly into two strands of literature. The first strand studies risk-sharing within the family and is primarily empirical in nature, which is also true of our approach. The empirical component of our paper corresponding to documenting facts about transfers builds on the work of McGarry (2016) and Andersen et al. (2020), who document that parents give transfers in response to adverse life events, including job losses, unexpected expenditures, and changes in family composition. We complement them by measuring how transfer size and composition vary across the wealth distribution.

Our analysis of the effects of parental insurance on children's decisions is also heavily

indebted to Boar (2021), which demonstrates that parents save to insure their children against permanent income risk. We expand on this work by considering the other side, children's responses rather than parents, in more detail. Our methods for doing so also attempt to capture the possibility that the transfers and insurance provided by parents may be jointly determined with children's choices.

The second strand studies the factors that contribute to the intergenerational persistence of income and wealth as well as the inequalities observed in them. This literature is primarily structural, using calibrated life-cycle models to account for the facts. This includes the foundational work of Huggett et al. (2011), which identifies initial variation in human capital as the crucial factor in explaining the dispersion of life outcomes. It also includes more recent papers, such as Abbott et al. (2019), Lee and Seshadri (2019), Boar and Lashkari (2021), Gallipoli et al. (2026), and Alon et al. (2026), which attempt, via comprehensive structural models, to obtain a deeper understanding of the sources of persistence and inequality, as well as the relationship between the two. Our paper is part of the same family, though it is unique for its detailed modeling of inter vivos transfers over the life cycle.

There are also a number of papers that study this issue using empirical methods which are complementary to the structural approach that we and others take. Charles and Hurst (2003) document the high degree of persistence in wealth levels, decomposing the persistence into a component explained by the persistence of permanent income and a component explained by the persistence of portfolio choices. Fagereng et al. (2021), mentioned earlier, use highly detailed Norwegian administrative data to study wealth persistence and its mediators. Focusing instead on permanent income, Lefgren et al. (2012) perform a decomposition of intergenerational persistence with restrictions based on Becker and Tomes (1979).

Layout. The paper is structured as follows. Section 2 establishes the aggregate facts regarding inter vivos transfers. Section 3 provides the micro-level evidence of the relationship between the amount of insurance provided by parents and the amount of risk taken on by children. Section 4 outlines the structural model we'll use to understand the macro-level effects of inter vivos transfers. Section 5 details the calibration procedure, the moments we aim to match, and the results of our counterfactuals. Section 6 concludes.

2 Facts about inter vivos transfers

In this section we describe the household survey data used throughout the paper and document several facts about inter vivos transfers from parent to child: (i) aggregate inter vivos transfers are economically large; (ii) transfers have a fat right tail and are disproportionately given by wealthy parents; (iii) the probability and size of transfers increase with job losses and other life events of the recipient, especially for wealthy parents; and (iv) transfers come disproportionately early in the recipients' life, and the type of transfers vary across the lifecycle.

2.1 Survey data

Health and Retirement Study. Our primary data on inter vivos transfers come from the Health and Retirement Study (HRS), a biennial panel survey detailing the health, wealth, income, and demographic characteristics of a representative sample of the U.S. population aged 50 and older. The original cohort, first surveyed in 1992, comprised individuals born between 1931 and 1941. Starting in 1998, the HRS began regularly adding new cohorts to preserve the survey's cross-sectional representativeness alongside its longitudinal structure. To avoid structural breaks associated with early changes in the survey design, our empirical analysis restricts the sample to the 19982022 waves.

In addition to detailed respondent-level data, the HRS collects broad demographic, employment, and income information about respondents' children, including those no longer living in the household. Crucially for our analysis, respondents report whether they made monetary transfers totaling at least $500 to each child since the previous interview, as well as the precise transfer amounts. We supplement this main sample with data from the 1992 and 1994 waves, which specifically query respondents on the total educational expenses incurred for each child. Finally, when a respondent passes away, the HRS conducts an exit interview with a proxy informant who knew the deceased well (typically a spouse, child, or other close family member). These exit surveys contain detailed questions regarding the allocation and exact values of any bequests, both monetary and real estate, and serve as our primary source for bequest data.

Panel Study of Income Dynamics. For supplementary data on transfers, income, and wealth we use the Panel Study of Income Dynamics (PSID). The PSID provides longitudinal information on a representative sample of U.S. households starting in 1968. The PSID began with an annual survey design, but starting in 1997 changed to conducting surveys every two years. Each survey provides granular income, wealth, demographic, and con-

sumption information. A key feature of the survey design, is that when children move out of an original sample PSID household they break off as their own sample unit and are thereafter surveyed in subsequent years. This structure allows one to form parent-child linkages, while obtaining detailed information on both. Moreover, the PSID asks recipients (donors) about the amount of transfers they receive (send) each survey. However the information is less detailed than that of the HRS; in particular, respondents are only asked about the total amount of transfers they receive or send from all family members. Nonetheless, as the PSID provides more detailed information about the recipient it serves as a useful supplement to our main empirical analysis in the paper. Due to changes in the survey design, a limited number of parent-child linkages in earlier years, and a desire to remain consistent with the HRS sample we focus on the 1999 to 2021 surveys.

2.2 Quantifying inter vivos transfers

We start by measuring aggregate yearly inter vivos transfers in the HRS. Due to the survey structure, we focus in particular on transfers from parent to child where the parent is at least 50 years old. We also exclude transfers in the form of tuition payments, which are explored further later in this section. Due to these caveats, along with potential under reporting bias, we interpret our estimates as a lower bound for the magnitude of inter vivos transfers.

Transfer Type	**Avg. Annual Transfer**	**Source**
Inter vivos (50 and older)	\$161B	HRS
Bequest (50 and older)	\$476B	HRS
Federal UI	\$78B	BEA
Federal Welfare	\$240B	BEA
All	*\$955B*	

Table 1: Annual Transfers, 1996-2019

Notes: This table reports the average annual transfers between 1996-2019. inter vivos transfers and bequests are calculated using the HRS, while Federal UI and welfare transfers come from the BEA. Direct educational payments are excluded from inter vivos transfers.

Table 1 reports the average yearly inter vivos transfers from 1996-2019[8] along with other commonly studied forms of transfers: bequests, federal UI payments, and federal welfare payments. For representativeness, inter vivos transfers and bequests are calcu-

[8]We exclude 2020 and 2021 data to avoid including extreme values associated with the Covid-19 period.

lated using survey weighted sums.[9] Federal UI and welfare data come from the Bureau of Economic Analysis (BEA). The table indicates that inter vivos transfers are smaller than bequests and federal welfare in aggregate, averaging roughly 1/3 and 2/3 the magnitude of each respective transfer type. On the other hand, they are more than double the size of federal unemployment insurance payments. Overall, total inter vivos transfers are economically large, and on a similar order of magnitude to other commonly studied forms of transfers.

Lifetime transfers. An object of interest is the lifetime transfers received (sent) by each recipient (donor). However, due to the limited sample length this object is not directly observable. For instance, a donor that is 50 years old in the 1998 wave and participates in every survey only reports transfers up to the age of 75. With this limitation in mind we calculate total per child inter vivos transfers between 1996 and 2021 at the donor level. Since we are interested in distributional features of these transfers, and thus ideally compare donors at similar ages, we restrict the sample to those that are between the ages of 50 and 55 in their first survey. Furthermore, we only consider households that participate in at least five surveys.[10] Figure 1 plots the distribution of total per child inter vivos transfers between 1996 and 2021. There is a large degree of heterogeneity. Approximately 30% of parents transferred nothing to their children, while 22% transferred at least $50,000 per child. Furthermore, over 2% of parents sent an astonishing $200,000+ per child. To capture the degree of upper tail inequality we estimate the Pareto coefficient of the top 10% using the Hill estimator:

$$\hat{\alpha}_{TR} = \left(\frac{1}{n} \sum_{i=1}^{n} \log \left(\frac{TR_i}{TR_{\min}} \right) \right)^{-1}, \tag{1}$$

where $i \in [1, n]$ are the set of donors with cumulative transfers in the 90th percentile, and $TR_{\min}$ is the lowest transfer amount of this set. The resulting Pareto coefficient is $\hat{\alpha}_{TR} = 2$, indicating that cumulative inter vivos transfers display a similar degree of concentration as labor income (Gallipoli et al., 2026) and slightly less concentration than bequests, which we estimate to be $\hat{\alpha}_{BEQ} = 1.46$.

While not possible at the donor level, average lifetime transfers can be estimated by combining the HRS data with mortality statistics. Using the pooled HRS panel from 1996-

[9] To adjust for missing bequest information, we rescale the survey weights to match the U.S. death distribution.

[10] In the case that a household misses one or more survey we re-normalize their cumulative transfers based on the number of missed observations.

Moment	Value
Mean	$33,836
Percentiles	
14.3%	$0
25%	$2,951
Median	$17,639
75%	$44,092
Share of all transfers	
Top 10%	43.0%
Top 5%	26.5%

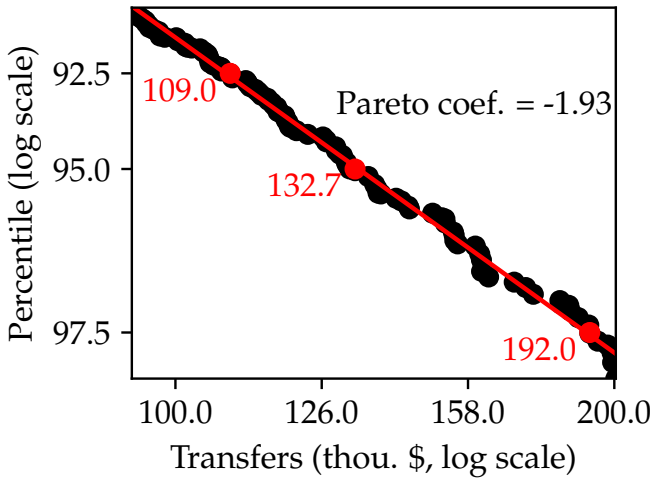


Figure 1: Distribution of cumulative transfers

Notes: This figure plots the donor-level distribution of total per child inter vivos transfers from 1996-2021.

2021 we calculate the average transfer amount received by children at each age. We then normalize each average using the cumulative mortality rate of the parents for a given cohort using the life tables provided by the U.S. Social Security Administration (SSA) to account for survivor bias in the sample. Overall, this exercise yields estimates of the expected per child transfer amount at each age conditional on the donor being at least 50 years of age.[11] Finally, we estimate direct educational transfers that took place before the survey years using the 1992-1994 supplementary data.

Given the evidence provided in Figure 1 that lifetime transfers are highly unequal, we repeat the analysis above separately for different parental wealth groups. We choose to focus on a measure of lifetime wealth, rather than contemporaneous wealth, to limit the causal effect of transfers on wealth. Our baseline measure of lifetime wealth is total wealth when the household head is age 65.[12] Finally, we split donors into four wealth groups based on wealth percentile: 0-25th, 25th-75th, 75th-95th, and >95th, representing the poor, middle, upper, and wealthy classes, respectively. Wealth groups are formed within birth cohorts, particularly those born in 1918-1927 (the Greatest Generation), 1928-1945 (the Silent Generation), and 1946-1965 (Baby Boomers).

We further decompose transfers into four broad categories. First, human capital trans-

[11] We do not account for potential differences in mortality rates between donors with different characteristics. Since wealthier individuals have higher life expectancy, this likely downward biases our results, particularly for top wealth groups.

[12] In cases where the head is not observed at age 65, we choose wealth at the closest age in the range of 55-75, dropping any households not observed in this window.

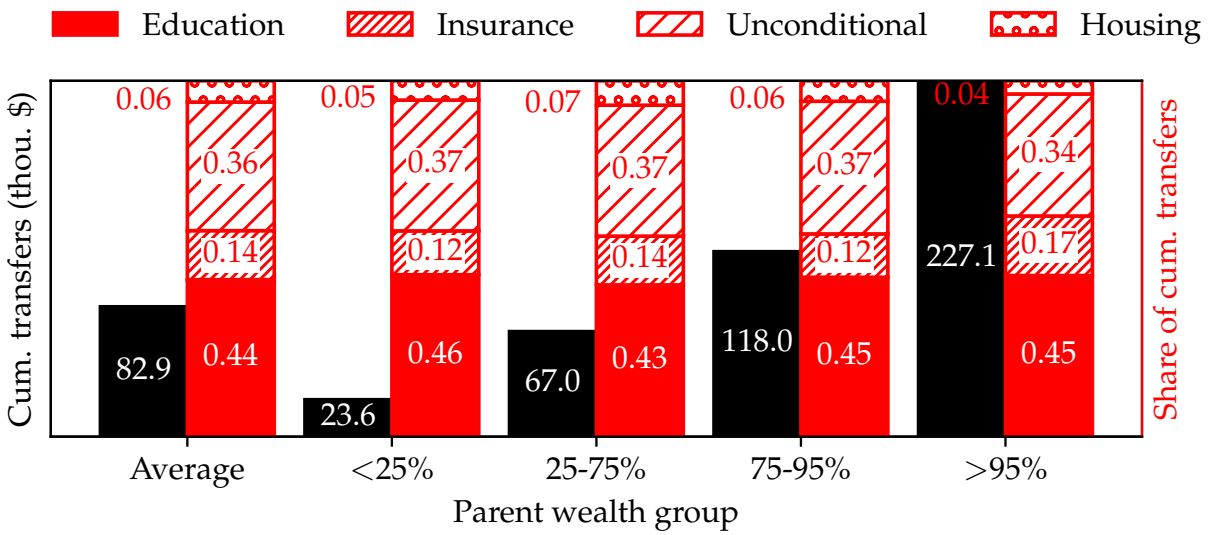


Figure 2: Average cumulative lifetime transfers share of total by type, by parent wealth group

Notes: This figure plots the estimated average lifetime transfers a child receives from their parents for the entire sample as well as broken out by lifetime donor wealth group. Shares of cumulative lifetime transfers are broken into fractions that went towards human capital, insurance of life events, home ownership, or miscellaneous (unconditional).

fers include direct education and school expense payments, as well as any monetary transfers made while the recipient is away at school. Second, insurance payments cover transfers tied to life events that reduce income or increase expenses, specifically: employment reductions (moving from full- to part-time, or becoming unemployed), marriage, divorce, or the birth of a child. Third, housing transfers include the value of directly transferred real estate and monetary support given when the child becomes a first-time homeowner.[13] Fourth, unconditional transfers comprise all remaining payments that do not fit into the previous three categories.

Figure 2 plots the average transfers received by children, both in aggregate, and for each of the four donor wealth groups. Average lifetime transfers are roughly $83,000, made up of 44% human capital transfers, 14% insurance transfers, 6% home transfers, and 36% unconditional transfers. The amount of transfers increases monotonically with each wealth group: the poor, middle, upper, and rich classes transfer approximately $24,000, $67,000, $118,000, and $227,000 to each child, respectively, illustrating that wealthier parents transfer more to their children. The composition of transfers also varies by group. For instance, the insurance category make up 17% of rich class' transfers but only 12% of the

[13]Because we cannot detect when a child moves between owned homes, we exclude transfers occurring during subsequent moves.

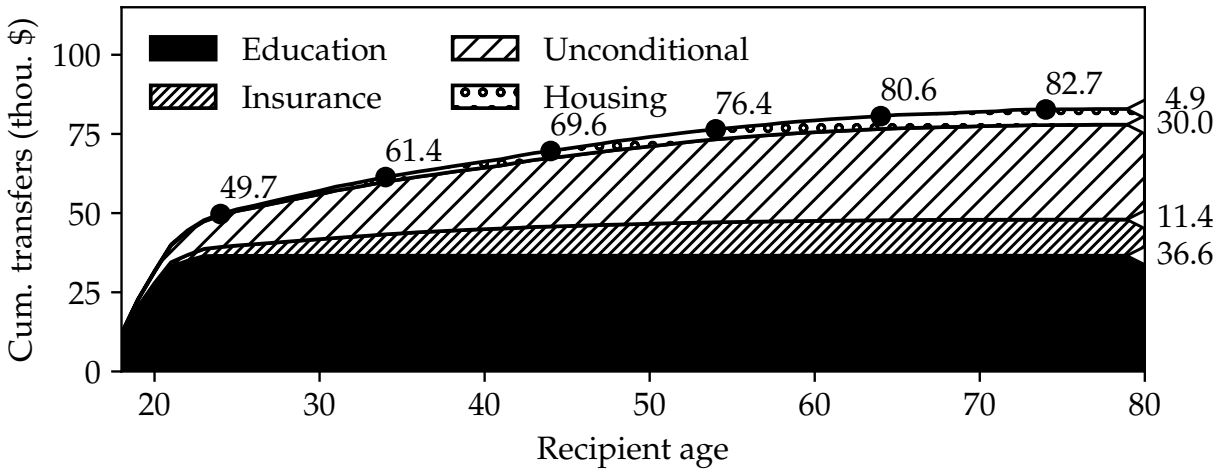


Figure 3: Average cumulative life cycle transfers by type

Notes: This figure plots the cumulative average lifetime transfers by recipient (child) age. Transfers are grouped by those that went towards human capital, insurance of life events, home ownership, or miscellaneous (unconditional).

poor class.[14] On the other hand, the rich class maintains the lowest proportion of transfers for home ownership of 4%. Interestingly, human capital transfers consistently make up between 43-46% of each wealth groups' lifetime transfers, suggesting that education is the dominant type of transfer for parents across the wealth distribution.

2.3 The timing of inter vivos transfers

We now turn our focus to the dynamics of inter vivos transfers. We broadly ask three questions: (i) how do the magnitudes and types of inter vivos transfers evolve over a recipients lifetime; (ii) to what extent do inter vivos transfers condition on recipient outcomes; and (iii) do these dynamics vary by donor wealth group.

Figure 3 plots the cumulative average transfers a child received from age 18 to 80. Strikingly, inter vivos transfers disproportionately come at the beginning of life: 69% of transfers come by the age of 30. Unsurprisingly, these front loaded transfers are dominated by human capital transfers, which tend to take place between the ages of 18 and 25. Conversely, just 11% of total transfers take place after age 50. While mortality rates explain some of this decline, other factorssuch as greater financial stability in middle agelikely play a role.

Given the skewed timing of the transfers an exercise of interest is to measure the

[14]It is worth noting that children of the rich experience far fewer insurance events, suggesting that these differences understate the difference in the degree of insurance present. We discuss this issue further in Section 2.3.

present value of the transfers. Intuitively, earlier transfers are more valued due to household impatience and their potential for real financial returns overtime. We specifically estimate the present value of the expected future lifetime transfers when the recipient is age 18 years old as:

$$PV_{TR}^{AVG} = \sum_{a=18}^{80} \frac{\mathbb{E}[TR_a]}{(1+r)^{a-18}},$$

where r is the real interest rate and TR_a is the transfer amount at age a. Table 2 reports the present values of expected inter vivos transfers and bequests for interest rates of 2%, 5%, and 7%. The first four rows report values for each of the wealth groups, while the last row reports values for the entire population. While lifetime aggregate bequests[15] are more than twice the size of aggregate inter vivos transfers, the aggregate present values tell a different story. At the lowest reported interest rate, the present value of bequests is only 24% higher than that of inter vivos transfers. Moreover, at the higher interest rates of 5% and 7%, the present value of inter vivos transfers is *larger* than the present value of bequests. Intuitively, because inter vivos transfers are more frontloaded than bequests, they are subject to a smaller discounting penalty at higher interest rates. The magnitude of inter vivos transfers relative to bequests also varies by wealth group. In particular, for the poor and middle classes, inter vivos transfers have a higher present value even at the lowest reported interest rate. Conversely, for the top 5%, the present value of bequests is approximately twice as large as the present value of inter vivos transfers at the same interest rate. This heterogeneity in relative present values is driven by the even heavier concentration of bequests at the top of the wealth distribution.[16] Nonetheless, for all wealth groups, inter vivos transfers maintain higher present values at interest rates of 5% and 7%. Overall, these results indicate that the economic importance of inter vivos transfers is consistently greater than raw nominal aggregates suggest.

The insurance channel. Beyond the mechanical value derived from their earlier timing, inter vivos transfers may provide additional utility because they can be strategically adjusted to align with the recipient's marginal needs. Figure 2 indicates that a small portion of transfers occurs during life events that likely lower the recipient's disposable income. However, this type of descriptive analysis does not identify whether transfers systematically increase during these periods. Moreover, because these life events are somewhat rare, the unconditional averages of their associated transfers may understate the role of transfers in insuring against changes in disposable income. To provide more causal evi-

[15] Lifecycle bequests are reported in Appendix Figure A.1

[16] Appendix Figure A.2 plots average lifecycle bequests by wealth group.

	r = 2%		r = 5%		r = 7%	
Bucket	PV^{AVG}_{IVT}	PV^{AVG}_{BEQ}	PV^{AVG}_{IVT}	PV^{AVG}_{BEQ}	PV^{AVG}_{IVT}	PV^{AVG}_{BEQ}
Bottom 25%	**$20,006**	$5,039	**$16,832**	$1,823	**$15,464**	$1,014
25 - 75%	**$56,238**	$38,447	**$47,226**	$14,490	**$43,508**	$8,355
75 - 95%	$97,892	**$139,483**	**$81,316**	$53,992	**$74,517**	$31,591
Top 5%	$187,822	**$373,458**	**$153,226**	$149,634	**$138,585**	$89,651
All	*$68,919*	***$85,715***	***$57,242***	*$32,017*	***$52,420***	*$18,353*

Table 2: Present discounted value of transfers by parent wealth group

Notes: This table reports the estimated present value of expected lifetime inter vivos transfers and bequests when the recipient is 18 years old. Values are reported separately by donor (parent) wealth group, and for different interest rates.

dence for this channel, we estimate the equation:

$$TR^{(g)}_{i,t} = \alpha + \beta \text{Event}_{i,t} + \sum_{a \in \{\text{Y,M}\}} \left[\gamma_a \text{Age}_{a,i,t} + \lambda_a (\text{Event}_{i,t} \times \text{Age}_{a,i,t}) \right] + \mathbf{X}'_{i,t}\boldsymbol{\theta} + \delta_t + \varepsilon_{i,t}, \tag{2}$$

where $TR^{(g)}_{i,t}$ denotes the transfer amount estimated for all recipients and separately across the four parental wealth groups (indexed by g), $\text{Event}_{i,t}$ represents a life event of the type described in section 2.2. The terms $a \in \{\text{Y,M}\}$ are indicator variables equal to 1 when the recipient is young (ages 18-30), and middle-aged (ages 31-54), respectively, $\mathbf{X}_{i,t}$ represents a vector of demographic controls including education, sex, marital status, and race, and δ_t captures year fixed effects. We note that this exercise is in a similar spirit to one conducted by McGarry (2016). The two main differences are that we allow for both heterogeneity by parental wealth and the age of the recipient.

Figure 4 reports the estimates over our sample, specifically for the event of a "job loss." The first column indicates that a recipient's job loss is associated with higher total transfers since the last interview, especially when the recipient is young, but also when they are middle-aged. Holding demographic factors and year fixed effects constant, the point estimates indicate that young recipients who lost their jobs between interviews received a predicted $2,809 per year: a 57% increase ($1,016 more) relative to the conditional baseline without a job loss. However, the magnitude of baseline transfers and the degree of adjustment vary considerably by donor wealth group. For example, a young recipient with parents in the bottom wealth group receives an expected $141 in yearly transfers when there is no job loss, a figure that does not significantly change upon job loss. Conversely, a young recipient with parents in the top wealth group receives a predicted baseline of

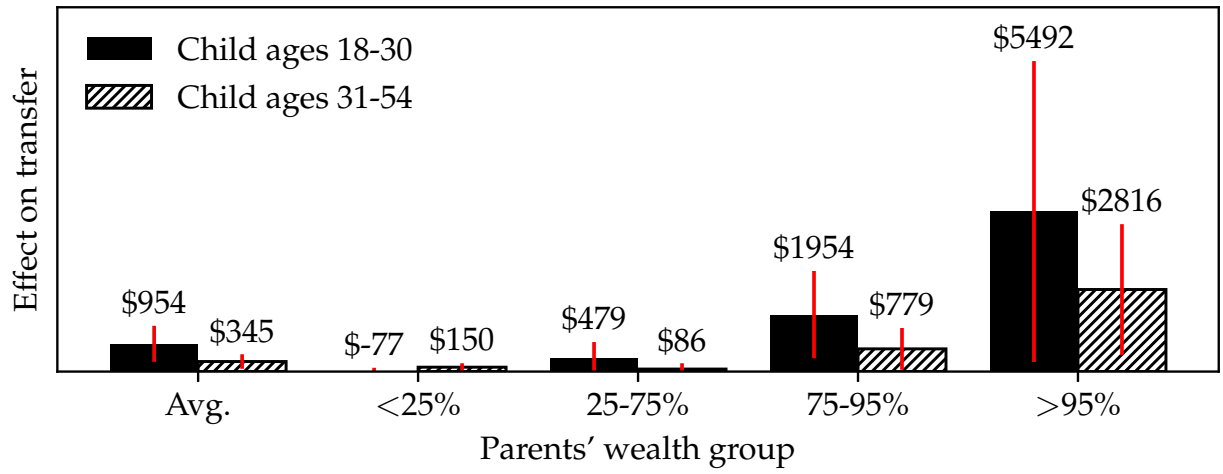


Figure 4: Job loss and transfers by age, by parent wealth group

Notes: This table reports the coefficients and 95% confidence intervals from estimating equation 2. The dependent variable is total transfer amount (2024 USD), and Job Loss is an indicator variable equal to 1 when the recipient changes job status from working to not working between consecutive surveys. The full table of regression coefficients can be found in Appendix Table **??**.

$6,905, which increases by 67% to $11,557 following a job loss. The two middle wealth groups exhibit baseline magnitudes and job-loss adjustments that fall between these two extremes. Appendix Tables A.1 and A.2 show that these patterns generally remain consistent for a larger set of life events in both the HRS and PSID, respectively. We interpret these estimates as lower bounds for the degree of adjustment to life events, since the data require us to average transfers over a two-year period, whereas the actual adjustment may occur over just a few months. Nonetheless, these results indicate that wealthier parents not only provide larger baseline transfers to their children, but they also adjust their transfers by a larger magnitude following a job loss.

3 Parental wealth and productive risk-taking

Having established in section 2 that wealthier parents provide a robust financial safety net through inter vivos transfers and informal insurance, this section explores how this backing alters children's behavior. We hypothesize that the presence of this safety net encourages children to take on greater risk in pursuit of higher expected returns. Because a strong wealth buffer cushions against severe income losses, it dampens precautionary savings motives, a mechanism consistent with the literature on conditional transfer programs (Hubbard et al., 1995; Engen and Gruber, 2001). Consequently, we document that children of wealthier parents take on significantly more productive risk across two key

dimensions: occupational choice and household portfolio allocation.

3.1 Job choice

We begin by analyzing the risk-reward profiles of the jobs chosen by children across the parental wealth distribution. To do so, we closely follow the methodology of Boar (2021) to measure a worker's expected permanent income and income uncertainty conditional on age and sector.

First, for an individual i at age a working in sector s, we model future income $y^i_{a',s}$ at any future age a' as:

$$y^i_{a',s} = \mathbb{E}_s[y^i_{a',s} \mid \mathcal{I}^i_{a,s}] + e^i_{a',a,s}, \tag{3}$$

where $\mathcal{I}^i_{a,s}$ represents the available information set at age a, and $e^i_{a',a,s}$ is the forecast error.

In the second step, we aggregate to the sector level to estimate expected permanent income, $Y_{a,s}$, and the standard deviation of the lifetime forecast error, $\sigma_{a,s}$. Their empirical estimators are constructed as follows:

$$\hat{Y}_{a,s} = \sum_{a'=a}^{A} \frac{\hat{y}_{a',a,s}}{R^{a'-a}}, \quad \hat{\sigma}_{a,s} = \left(\sum_{a'=a+1}^{A} \frac{\widehat{\mathrm{Var}}_s(e_{a',a,s})}{R^{2(a'-a)}} + 2 \sum_{a'=a+1}^{A-1} \frac{1}{R^{a'-a}} \sum_{k=a'+1}^{A} \frac{\widehat{\mathrm{Cov}}_s(e_{a',a,s}, e_{k,a,s})}{R^{k-a}} \right)^{\frac{1}{2}},$$

where A is the maximum working age, and R is the gross discount rate. The term $\hat{Y}_{a,s}$ represents the discounted sum of estimated expected future income, while $\hat{\sigma}_{a,s}$ is the estimated standard deviation of the discounted sum of forecast errors.

Data and estimation details. We assign each household head to one of 25 sectors defined by the head's industry and occupation on the main job, following and extending the classification in Boar (2021). The key difference is our treatment of the self-employed. Self-employed workers exhibit wage profiles that differ markedly from those of wage and salary workers in the same industry and occupation, so pooling the blurs the earnings paths that define each sector. Rather than placing heads who are self-employed on their main job in the same industryoccupation cells as wage and salary workers, we group them into four separate self-employed sectors by industry: agriculture, mining and construction; goods production and distribution; business, personal and entertainment services; and finance, insurance, real estate and professional services. appendix B.1 describes the industry and occupation groups and how we harmonize the PSID's occupation and industry codes across survey years.

Estimating equation 3 requires specifying both the worker's information set, $\mathcal{I}^i_{a,s}$, and

the estimation procedure for the conditional expectation. The information set $\mathcal{I}^i_{a,s}$ includes the worker's current and lagged log labor earnings, education (high school or less, some college, college degree), race, marital status, household size, a quadratic in age, and a linear calendar-year trend for the year they reach age a'. We approximate the conditional expectation $\mathbb{E}_s[\cdot \mid \mathcal{I}^i_{a,s}]$ via an OLS projection of log earnings at target age a' onto these covariates, estimated separately for each sector s and target age by pooling all workers in sector s at ages $a < a'$. Sector affiliation is fixed at the forecast date, meaning workers who subsequently switch sectors still contribute their realized age-a' earnings to the estimation of their original sector. Finally, we convert the fitted log earnings to levels, $\hat{y}^i_{a',s} = \exp\left(\hat{\mathbb{E}}_s[\log y^i_{a'} \mid \mathcal{I}^i_{a,s}]\right)$, to recover the forecast error $e^i_{a',a,s} = y^i_{a',s} - \hat{y}^i_{a',s}$.

Our sample consists of household heads aged 22 to 65 in the original PSID core sample, observed from 1968 through 2019. We exclude the 2021 and subsequent waves to avoid abnormal wage dynamics during the COVID-19 pandemic and recovery. When estimating equation 3, we retain heads who are either working or unemployed, ensuring that transitions into unemployment contribute to measured income uncertainty. For the subsequent analysis, however, we restrict the sample to employed heads who can be assigned to a specific sector. To address the PSID's shift to biennial interviews after 1997, we construct a continuous annual earnings panel by combining each wave's report of the previous year's labor earnings with retrospective reports of earnings two years prior, filling the gaps between survey years. Finally, we set R = 2%. Appendix B.1 details the remaining sample restrictions and the construction of our earnings measure.

Parental wealth and wage profiles. We now ask how parental wealth shapes the expected permanent income and income uncertainty of the sectors children select to work in. In doing so, we face two main empirical challenges. The first is reverse causality: as argued by Boar (2021), parents may adjust their savings behavior in response to their children's income profiles. Thus, a child's choice of sector can directly impact measured parental wealth. The second challenge is isolating our mechanism of interest. While we posit that parental wealth influences sectoral choice via the safety net channel established in section 2, wealth may proxy for other confounding factors. For instance, wealthier parents may simply transmit a higher baseline risk tolerance to their children through their upbringing. Our empirical strategy is designed to overcome these endogeneity concerns.

We estimate equations of the form:

$$\log(Y_{i,t}) = \beta_g + \mathbf{X}'_{i,t}\boldsymbol{\theta} + \delta_t + \varepsilon_{i,t}, \tag{4}$$

$$\log(\sigma_{i,t}) = \beta_g + \gamma \log(Y_{i,t}) + \mathbf{X}'_{i,t}\boldsymbol{\theta} + \delta_t + \varepsilon_{i,t}, \tag{5}$$

where β_g is a fixed effect for parental wealth group g, $\mathbf{X}_{i,t}$ is a vector of child and parental controls, and δ_t are time fixed effects. In the uncertainty regression, we explicitly control for the child's expected permanent income, $\log(Y_{i,t})$, so that our estimates capture relative income uncertainty, isolating the effect of parental wealth on risk-taking conditional on a given expected reward. To address reverse causality, we instrument the child's lifetime parental wealth group, g, using measures of parental wealth during the child's youth. Specifically, we target the parent's log wealth and wealth group indicators measured when the child was 16 years old.[17] The identifying assumption is that parental wealth at this stage is not influenced by the child's future occupational choice. Intuitively, at age 16, children typically have not yet determined their permanent sector of employment; even if they have, parents are unlikely to have already internalized this future choice into their own wealth accumulation. The vector $\mathbf{X}_{i,t}$ includes the baseline demographic controls from equation 3 measured for both the child and the head parent, alongside age fixed effects and the parent's log expected permanent income and income risk (which are instrumented analogously). These controls are designed to absorb omitted confounding factors, such as the intergenerational transmission of human capital or preferences, that correlate with both parental wealth and the child's career path. We estimate Equations 4 and 5 via two-stage least squares (2SLS) using a sample of child-parent pairs from 1999 to 2023, aligning with the introduction of regular PSID wealth modules.[18] Appendix B.2 provides details on the child-parent matching and sample restrictions.

Figure 5 plots the estimated effects of parental wealth on children's expected permanent income and relative income uncertainty, expressed as percentage deviations from the bottom parental wealth group. The left panel demonstrates that, conditional on education and other covariates, children of wealthier parents select into sectors with higher permanent income. Specifically, those from the top two wealth groups choose career paths offering approximately 10% higher permanent income. The right panel reveals a parallel pattern for relative risk. Children with parents in the 75-95% wealth group select into sectors with 8.3% higher income uncertainty, and this effect rises to 15.5% for those with parents in the top 5%.

Overall, these findings are consistent with the hypothesis that parental wealth acts as a financial backstop, encouraging children to pursue higher-risk, higher-reward career paths. While one might attribute this increased risk tolerance to the anticipation of larger future bequests, occupational choices are primarily made early in the life cycle when the

[17] If parental wealth is not observed precisely at age 16, we use the closest available observation between ages 10 and 22. Wealth group cutoffs are constructed by child cohort.

[18] In all specifications, the first-stage F-statistics comfortably exceed standard thresholds for weak instruments.

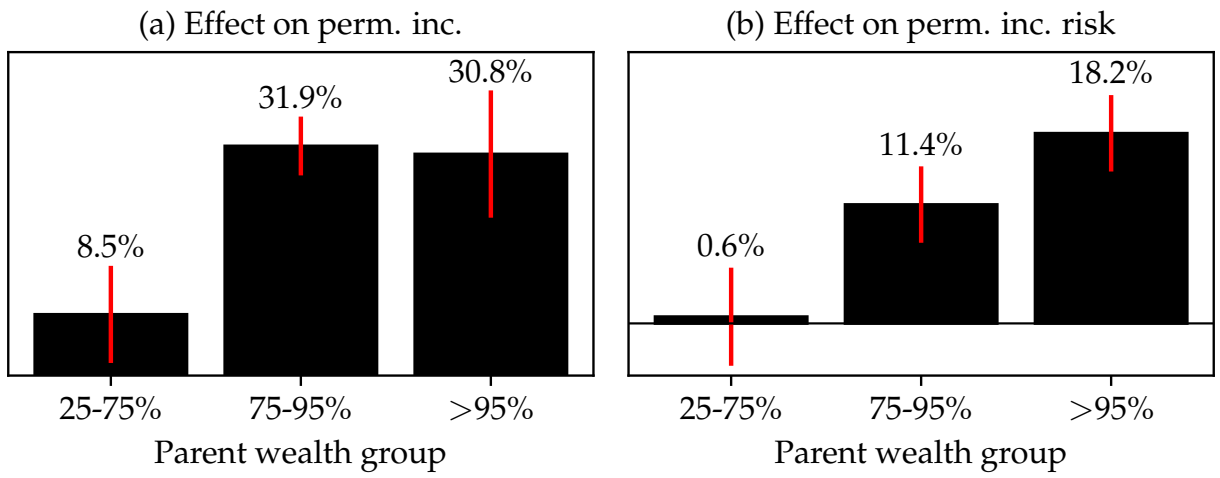


Figure 5: Parental wealth effect on income profile

Notes: This figure plots the estimated effects of parental wealth on children's expected permanent income (left panel) and relative income uncertainty (right panel). Estimates are interpreted as the percentage increase relative to the omitted bottom parental wealth group. Red bands represent 95 percent confidence intervals.

timing of a bequest is highly uncertain and realized inheritances are often decades away. A more plausible explanation is that wealthier parents offer a reliable safety net during the child's prime working years. This can operate through unconditional transfers that establish a baseline wealth buffer or through the implicit promise of informal insurance against severe downside income shocks.

3.2 Asset portfolios and savings

Using the same empirical strategy from our analysis of occupational choice, we next examine how parental wealth influences children's asset allocation and savings decisions. Specifically, we focus on three key variables: the fraction of wealth invested in risky assets, business ownership, and the overall savings rate.

Measurement. To capture both portfolio risk and precautionary savings motives, we construct three dependent variables using the biennial PSID wealth modules. First, we define the risky asset share as the ratio of risky assets to total wealth. Our definition of risky assets includes stocks, mutual funds, investment trusts, and business or farm equity. The remaining assets comprising total wealth are checking and savings accounts, money market funds, certificates of deposit, savings bonds, Treasury bills, IRAs and annuities, real estate, and vehicles. Second, we measure business ownership using an indicator variable equal to one if the household "owned a business at any time last year, or had a fi-

nancial interest in any business enterprise." Finally, we define the household savings rate as $\frac{y_{i,t}-c_{i,t}}{y_{i,t}}$, where $y_{i,t}$ is after-tax household income and $c_{i,t}$ is total household consumption. To estimate federal and state income taxes, we use the NBER TAXSIM program. Because the PSID expanded its expenditure coverage in 2005, we follow Attanasio and Pistaferri (2014) to impute missing consumption categories for the 1999, 2001, and 2003 survey waves. Appendix B.2 provides further details on the assets, business ownership measure, and consumption imputation.

Parent wealth, asset allocation, and savings. To measure the effect of parental wealth on asset allocation and savings rates we estimate equations of the form:

$$a_{i,t} = \beta_g + \mathbf{X}'_{i,t}\boldsymbol{\theta} + \delta_t + \varepsilon_{i,t}, \tag{6}$$

where $a_{i,t}$ represents the risky asset share, business ownership indicator, or savings rate. The vector $\mathbf{X}_{i,t}$ contains the baseline child and parent demographic covariates introduced in our occupational choice analysis. To ensure we are comparing children facing similar observable financial constraints, we additionally control for the child's own wealth group and income quintile. This allows us to isolate the marginal effect of having wealthier parents, conditional on the child's own financial resources. Furthermore, to rule out confounding channels such as the intergenerational transmission of risk preferences, imitation, or financial literacy, we control for the head parent's analogue to the dependent variable in each specification (e.g., controlling for the parent's risky asset share when analyzing the child's risky asset share). Following our established empirical design, we estimate equation 6 via two-stage least squares (2SLS) on the 19992023 child-parent sample. We instrument the lifetime parental wealth group, alongside these new parental behavioral controls, using their respective values observed when the child was 16 years old.

Figure 6 plots the estimated effects of parental wealth on portfolio composition and savings behavior, expressed as percentage point deviations from the bottom parental wealth group. Conditional on the child's own wealth, income, and other covariates, the left and middle panels demonstrate that a stronger parental financial background drives children to take on more portfolio risk. Specifically, having parents in the top two wealth groups significantly increases both the fraction of wealth allocated to risky assets and the probability of owning a business. These effects are particularly pronounced at the top of the distribution. An assignment to the top 5% of parental wealth increases a child's risky asset share by 14.8 percentage points and raises their likelihood of owning a business by 9.4 percentage points. The estimates are small and statistically insignificant for the 25-75%

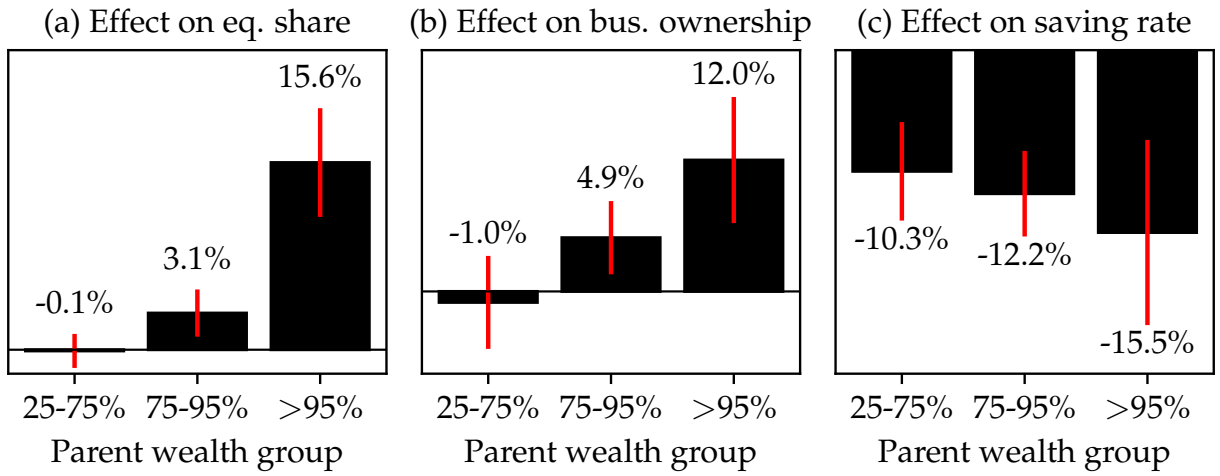


Figure 6: Parental wealth effect on asset portfolios and savings

Notes: This figure plots the 2SLS estimates of the effect of parental wealth on children's risky asset share (left panel), business ownership probability (middle panel), and savings rate (right panel). Point estimates represent the percentage point difference relative to the omitted bottom parental wealth group. Red bands indicate 95 percent confidence intervals.

group, suggesting substantial nonlinearities in how parental wealth facilitates financial risk-taking. Finally, the right panel reveals a monotonically decreasing effect of parental wealth on the savings rate. Relative to the bottom baseline group, having parents in the 25-75%, 75-95%, and top 5% groups reduces the child's savings rate by 6.9, 14.7, and 19.6 percentage points, respectively.

These financial outcomes closely mirror our findings on career trajectories, reinforcing a consistent behavioral response. The presence of a robust parental wealth buffer systematically reduces the need for precautionary savings and emboldens children to take on more productive risk across all major dimensions of household economic decision-making. To formally quantify this safety net and explore its aggregate implications, in section 4 we develop a structural life-cycle model. While our theoretical framework abstracts from explicit portfolio allocation and discrete occupational variance, it captures the core mechanism of this risk-taking channel through the lens of human capital accumulation. Specifically, the availability of parental insurance allows children to forgo early precautionary savings and instead dedicate more time and resources to building human capital. This modeling approach elegantly encapsulates the fundamental trade-off documented in our empirical results: a reliable parental backstop crowds out the need for self-insurance, freeing the child to undertake investments with higher expected lifetime returns.

4 Modeling the effects of transfers

We develop a discrete-time overlapping generations (OLG) economy with endogenous human capital accumulation, idiosyncratic income risk, incomplete markets, and borrowing constraints.[19] We pursue the added complication of college attendance decisions to add a costly extensive margin to human capital production, which complements the intensive margin of the accumulation which occurs on the job. Crucially, for our purposes, we also allow for multidimensional heterogeneity in intergenerational transfers from parents to children.

Each period in the model is a year. Initially, agents choose whether to attend college to build human capital or enter work immediately. Once they're in the labor force, they consume, save, work, and learn on the job. While they're working, they may receive some transfers from their parents for a fixed number of years. After that, they may give transfers to their own children for the same number of years, which may extend into retirement. After working, they retire and receive social security, choosing consumption and saving, while taking the transfers they give to their children as given. Life ends after retirement.

4.1 The college decision

Agents enter the model at age 18 and must immediately make a discrete choice regarding college attendance. If they forgo college, they proceed directly to their working lives equipped with their initial level of human capital and learning ability.

The cost of college. Each year of college costs, in total, P dollars, which includes all the expenses involved in schooling. For simplicity, we assume no interest accrues while in school,[20] so the total principal outstanding at the time of graduation is PT_c, where T_c denotes the number of years spent in college. We assume that the balance of the loan is paid in constant installments over a fixed term T_{sl} and that the remaining balance accumulates interest at the rate R_{sl}. Then, the yearly payments are equal to

$$p_{sl} = \left(1 - \frac{1}{R_{sl}}\right)\left(1 + \frac{1}{R_{sl}^{T_{sl}} - 1}\right) PT_c. \tag{7}$$

[19] The model in which we study the effects of IVTs is similar to the now workhorse model of Huggett et al. (2011) and the richly extended version of it in Alon et al. (2026), though we pursue a more detailed model of transfers during adult hood than they do.

[20] This is the case with federal subsidized loans, but other types of loans accrue interest during schooling. If one desires to add it, it could be be folded into the cost P or added directly in the formula for the principal.

Life at college. A fraction θ of the cost of college P represents the costs that are associated with agents' consumption while away at school, such as the cost of room and board as well as various amenities offered by the college. Hence, for each of the T_c years at school, agents receive endowments of the single consumption good equal to θP. We make the additional assumptions that this endowment is non-transferable and cannot be stored across periods.

The remaining fraction $1 - \theta$ of college costs consists of the payments, part of tuition and fees, that go toward the costs of human capital accumulation but lack direct consumption value. Then, paying that cost, agents that attend college produce human capital for the next year according to the production function

$$h' = F(h, i, z) = z(hi)^{\gamma} + (1 - \phi)h, \tag{8}$$

where h is the starting level of human capital, i is the fraction of the agent's time allocated to learning, and z is the agent's learning ability. While in college, we assume that $i = 1$, so that all the time which could be allocated to productive uses during one's college years is allocated to learning. A student that matriculates with human capital h and learning rate z would therefore graduate with human capital equal to $F^{(\circ T_c)}(h, 1, z)$, where $F^{(\circ T_c)}$ denotes T_c applications of the function $F(\cdot, 1, z)$ to the input level of human capital. Similarly, we may also define the growth rate in human capital while at college as $g(h, z) = F^{(\circ T_c)}(h, 1, z)/h$.[21]

On the interpretation of these initial state variables at age 18, it's natural for us to think of earlier investments by parents as primarily impacting the learning rate z rather than the level of human capital h. This is primarily because we regard these investments as complementary to college education, increasing its returns as well as those of subsequent investments of agents' time, which is true of an increased learning rate z. On the other hand, we might think of the starting level of human capital h as a bundle of skills with some level of marketability if one were to join the labor force immediately, which implies those with more human capital at the end of high school may find college a less attractive option, though this is not always the case. We have in mind here, for example, students with exceptional programming abilities as well as students that spent time in vocational or trade programs in high school.

Deciding on college. The decision to attend college depends on the agent's initial human capital h and learning ability z, any college wage premia, and the agent's expectations

[21] Under the assumption that $F(\cdot)$ takes the form in (8) with $\gamma \in (0, 1)$, it is straightforward to show that $\partial_h g(h, z) < 0$ and $\partial_z g(h, z) > 0$. In other words, the potential returns to college education right out of college are decreasing in the starting level of human capital and increasing in the learning rate.

about what transfers they'll receive from their parents and what transfers they expect to give to their own children. In our analysis, we allow parents to make children's transfer types depend on children's college attendance.

After graduating from college, an agent starts work with zero assets, human capital equal to $g(h,z)h$, and idiosyncratic labor productivity ϵ. The value of being at this particular state is $V_{c,0}(0, g(h,z)h, \epsilon; z, \omega_c)$, taking as given the agent's learning ability z and transfer type ω_c. The transfer type ω_c is a collection of objects that defines the agent's parents' transfer function, their own expected transfer function once they have children, and any payments they'll be required to make on their student loans after they graduate. It captures the lump sum payments the agent will make and receive, as well as the degree of insurance they'll receive from their parents.

We evaluate expected lifetime utility using recursive Epstein-Zin preferences, which allow us to separate the elasticity of intertemporal substitution (EIS), denoted by η_c, from the coefficient of relative risk aversion (CRRA), denoted by η_r. The expected lifetime value of attending college, evaluated at age 18, is

$$W_c(h,z;\omega_c) = \left(\sum_{t=0}^{T_c-1} \beta^t (\theta P)^{\frac{\eta_c - 1}{\eta_c}} + \beta^{T_c} \left(\mathbb{E}_\epsilon \left[V_{c,0}(0, g(h,z)h, \epsilon; z, \omega_c)^{1-\eta_r} \right]^{\frac{1}{1-\eta_r}} \right)^{\frac{\eta_c - 1}{\eta_c}} \right)^{\frac{\eta_c}{\eta_c - 1}} \tag{9}$$

which includes the value of consumption while at college and the certainty equivalent of the agent's value once they graduate college. If the child does not go to college, their expected value is

$$W_n(h,z;\omega_n) = \mathbb{E}_\epsilon \left[V_{n,0}(0, h, \epsilon; z, \omega_n)^{1-\eta_r} \right]^{\frac{1}{1-\eta_r}}, \tag{10}$$

which is the certainty equivalent of the value of starting work immediately, where ω_n is the agent's transfer type if they don't go to college.

Finally, the agent chooses whether to go to college by solving the discrete choice problem

$$\max_{j \in \{c,n\}} W_j(h,z;\omega_j) + \nu_j, \tag{11}$$

where $j = c$ denotes college attendance, $j = n$ denotes immediate labor force entry, and ν_j is an i.i.d. Gumbel (GEV1) taste shock with scale parameter σ. Under these assumptions, the probability that an agent goes to college is a function of h, z, ω_c, and ω_n, which takes

a standard form, provided in Appendix D.3.

4.2 Working lives

Upon entering the labor force, agents take their learning ability and transfer type as given. Further, as described above, the length of the work life of an agent depends on their type. The number of periods an agent of type j works before retiring is $T_{w,j}$, which is equal to $T_w - T_c$ for college attendees ($j = c$) and T_w for non-attendees ($j = n$).

We assume that an agent is endowed with a single unit of time. The agent chooses a fraction i of their time to spend building new human capital, which is produced according to the production function $F(h, i, \delta^t z)$, defined in (8). Here $\delta^t z$ reflects the change in the agent's learning ability over time, δ is the rate at which the agent's learning ability decreases while in the labor force and t is the number of periods since entry. This captures the potential difficulty of retraining once job or firm-specific human capital has been accumulated.

The remaining fraction of agents' time $1 - i$ is supplied inelastically in the labor market. Agents do not experience any disutility from working, so the trade-off between working and learning comes only from trading current income for future income.

For each unit of time supplied in the labor market, the agent earns $wh \exp(\epsilon)$ units of income, where w is the equilibrium wage rate. As is standard in the literature (Huggett et al., 2011), the idiosyncratic labor productivity shock ϵ is drawn from a Normal distribution, $\epsilon \sim N(\mu_\epsilon, \sigma_\epsilon^2)$, so that the productivity multiplier $\exp(\epsilon)$ is log-normal.

Depending on their transfer type ω and productivity realization ϵ, agents may receive an insurance transfer from their parents, equal to $whf_t(\epsilon, \omega)$, where $f_t(\epsilon, \omega)$ is a function that defines the size of the shock-dependent payment. Similarly, $\tau_{j,t}(\omega)$ is a type-specific sequence of net unconditional transfers given or received by the agent over their life. The specific forms of these functions and their relationship to the transfer type ω are discussed in more detail later in the section.

The final source of income is capital income. An agent endowed with a units of a risk-free one-period asset receives Ra units of income, where R is the gross risk-free rate. Then, a portion of all the income received by the agent may be used to purchase consumption c or a' units of the the risk-free asset. We assume that agents cannot borrow, so we impose the constraint $a' \geq 0$.

For the all the periods $t \in \{0, \ldots, T_{w,j} - 2\}$ prior to the final year of working life, agents'

value functions satisfy

$$
\begin{aligned}
V_{j,t}(a,h,\epsilon;z,\omega) = \max_{c,a',i,h'} & \left(c^{\frac{\eta_c-1}{\eta_c}} + \beta \left(\mathbb{E}_{\epsilon'} \left[V_{j,t+1}(a',h',\epsilon';z,\omega)^{1-\eta_r} \right]^{\frac{1}{1-\eta_r}} \right)^{\frac{\eta_c-1}{\eta_c}} \right)^{\frac{\eta_c}{\eta_c-1}} \\
a' + c &\leq Ra + (1-\mathcal{T}_\ell)\, wh \exp(\epsilon)(1-i) + whf_t(\epsilon,\omega) + \tau_{j,t}(\omega) \\
h' &= \delta^t z(hi)^\gamma + (1-\phi)h \\
a' &\geq 0;\ i \in [0,1].
\end{aligned}
\tag{12}
$$

For $T_{w,j} - 1$, the last period before retirement, we assume that agents no longer invest in human capital (i=0) and instead use their full unit of time to work. Since they only make a decision about the amount of consumption and saving, the value functions in this period instead satisfy

$$
\begin{aligned}
V_{j,(T_{w,j}-1)}(a,h,\epsilon;z,\omega) = \max_{c,a'} & \left(c^{\frac{\eta_c-1}{\eta_c}} + \beta V_{j,T_{w,j}}(a',h;\omega)^{\frac{\eta_c-1}{\eta_c}} \right)^{\frac{\eta_c}{\eta_c-1}} \\
a' + c &\leq Ra + (1-\mathcal{T}_\ell)\, wh \exp(\epsilon) + wh\, f_{(T_{w,j}-1)}(\epsilon,\omega) + \tau_{j,(T_{w,j}-1)}(\omega) \\
a' &\geq 0.
\end{aligned}
\tag{13}
$$

4.3 Retirement

During retirement periods, agents receive a constant stream of income from social security $\mathcal{S}(wh)$. Similar to the approach in Lee and Seshadri (2019), we assume this income is a function of agents' base income before retirement.[22] As before retirement, agents also receive net unconditional transfers and capital income. This income is either consumed or saved, with asset choices subject to the borrowing constraint. Then, for periods $t \in$

[22] This is necessary to economize on the number of state variables. If we were to use a formula for social security payments closer to that in reality, it would be necessary to include the average lifetime income as a state variable. In practice, however, appropriately chosen functions of wh can approximate the average income reasonably well. Like Lee and Seshadri (2019), we assume that $\mathcal{S}$ is a linear function so $\mathcal{S}(x) = s_0 + s_1 x$.

$\{T_{w,j}, \ldots, T_{\ell,j} - 2\}$,[23] agents' value functions are given by

$$\begin{aligned} V_{j,t}(a,h;\omega) = \max_{c,a'} & \left(c^{\frac{\eta_c - 1}{\eta_c}} + \beta V_{j,t+1}(a',h;\omega)^{\frac{\eta_c - 1}{\eta_c}} \right)^{\frac{\eta_c}{\eta_c - 1}} \\ a' + c &\leq Ra + \mathcal{S}(wh) + \tau_{j,t}(\omega) \\ a' &\geq 0, \end{aligned} \tag{14}$$

while in the final period of life $t = T_{\ell,j} - 1$, agents' choose $a' = 0$ and set $c = Ra + \mathcal{S}(wh) + \tau_{j,t}(\omega) \geq 0$, so

$$V_{j,T_\ell - 1}(a,h;\omega) = Ra + \mathcal{S}(wh) + \tau_{j,t}(\omega). \tag{15}$$

The net transfer term $\tau_{j,t}(\omega)$ includes bequests left to children, which enter as a negative value on the income side of the budget constraint.

4.4 Transfer types and transfer functions

As outlined above, the transfer type ω_j, which depends on the agent's education status, captures the sequence of net unconditional transfers the agent will give or receive, as well as the sequence of potential state-dependent transfers they might receive.

Insurance transfers and insurance policies. On the recipients' side, the children's side, the function $f_t(x,\omega)$, which defines the replacement rate in terms of the realization of $\epsilon = x$, is a function of three parameters, which are indexed by the period t, education level j,[24] and the transfer type ω. The precise form we use is

$$f_{j,t}(x,\omega) = \alpha_{0,j,t}(\omega)\,\Gamma\left(\alpha_{1,j,t}(\omega)x + \alpha_{2,j,t}(\omega)\right)\left(\mathbb{E}_\epsilon[\exp(\epsilon)] - \exp(x)\right) \tag{16}$$

The first term $\alpha_{0,t}(\omega)\,\Gamma\left(\alpha_{1,t}(\omega)x + \alpha_{2,t}(\omega)\right)$ determines the replacement rate, the fraction of the shortfall in idiosyncratic productivity $\mathbb{E}_\epsilon[\exp(\epsilon)] - \exp(x)$ that parents will replace when $\exp(x)$ is below its average.

Within the replacement rate term, $\Gamma(\cdot)$ is a CDF, α_0 is the maximum replacement rate, α_1 captures the how the replacement rate varies with the realization of ϵ, and α_2 is a shifter

[23] As with working life, the life span is measured from entry into the labor force, so $T_{\ell,j} = T_\ell$ for non-attendees ($j = n$) and $T_\ell - T_c$ for college graduates ($j = c$).

[24] Though this is only to control the number of periods for which there is a non-zero replacement rate, rather than having a direct dependence on education level.

that effectively determines the states in which an agent receives an insurance transfer.[25] In our exercise, we assume that only the replacement rate $a_{0,t}(\omega)$ varies with time. The sequences of parameters that determine the situations in which a child will receive a transfer $\{\alpha_{1,t}(\omega), \alpha_{2,t}(\omega)\}$ are fixed in time.

Requiring parents to continuously track the child's level of human capital and idiosyncratic productivity increases the size of the parents' state space greatly, making the model far less tractable. Therefore, we assume that, instead paying for all the insurance transfers directly, parents buy an insurance policy with payments included into the fixed type-specific sequence of transfers $\{\tau_{g,t}(\omega)\}_t$ they give to their children. We discuss this sequence in more detail below.

Educational and unconditional transfers received. Parents pay a fraction $e(\omega_c)$ of their child's educational costs, which are remitted directly to the school. Therefore, the school receives T_c payments of $e(\omega_c)P$ from the parents and the child makes T_{sl} annual payments of $(1 - e(\omega_c))p_{sl}$ after graduating.

After graduation, children receive a fixed sequence of payments $\{\tau_{u,t}(\omega_c)\}_{t=0}^{T-T_c-1}$. If the child, instead, goes immediately into the workforce, they receive the sequence $\{\tau_{u,t}(\omega_n)\}_{t=0}^{T-1}$. Notice that this sequence of transfers, like insurance transfers, depends on time. We choose this sequence to be decreasing in time, at a rate which broadly matches the time path of average unconditional transfers given.

Transfers given and the net transfer sequence. The sequence of transfers given by parents $\tau_{g,t}(\omega)$ is, conditional on transfer type ω, a deterministic sequence of transfers made to children. We choose the parameters of these sequences to approximate the average empirical paths of transfers that children receive over their lives.[26]

It is important to note that parents do not explicitly optimize over the timing of these transfers. Rather, the timing is calibrated to reflect the realized transfer paths in the data. Moreover, conditional on ω, parents take their sequence of outbound transfers as given. However, we interpret these sequences as reflecting the outcome of a prior discrete choice problem over possible transfer paths, where the timing is selected beforehand with the children's best interests in mind.

[25] In essence, this is a smooth version of an insurance policy that yields a fixed replacement rate if the realization for productivity is far enough below the average. That is, the CDF is effectively 1 for a very bad shock and effectively 0 for a very good shock.

[26] Parents save to make the transfers, some of which will be used to insure children, meaning we capture a part of the dynastic precautionary savings motive. However because they are deterministic, not responding directly to children's idiosyncratic states, we don't entirely capture the precautionary savings in Boar (2021) that generate an increasing path of consumption post-retirement.

These simplifications are appropriate because our focus is primarily on the children's side. That is: we concentrate more on how the timing and mix of transfers we observe in the data affect children's outcomes, rather than providing a detailed account of the exact mechanisms driving parental transfer decisions. Though, we want to emphasize the fact that the model is not silent on parents' motivations, since we are able to back out the way parents value their children's welfare from our modeling assumptions and the data.

Finally, having defined all lump-sum transfers given and received, we define the net transfer function over an agent's entire life $\tau_{j,t}(\omega)$, as follows:

$$\tau_{j,t}(\omega) = \begin{cases} \tau_{u,t}(\omega); & (t < T) \wedge (j = n) \\ \tau_{u,t}(\omega) - (1 - e(\omega))p_{sl}; & (t < T_{sl} < T - T_c) \wedge (j = c) \\ \tau_{u,t}(\omega); & (T_{sl} \leq t < T - T_c) \wedge (j = c) \\ -\tau_{g,j,t}(\omega); & ((T \leq t) \wedge (j = n)) \vee ((T - T_c \leq t) \wedge (j = c) \end{cases} . \tag{17}$$

4.5 Closing the model

Having described the households in the model, we turn to the remaining participants in the economy, the firms and the government, as well as the market-clearing conditions that link them together with the households. We specify all of the components under the assumption of a stationary equilibrium.

Firms. Firms produce output using a production function that exhibits constant returns-to-scale in capital and labor. They take wages w and interest rates r as given. Firms rent capital K from the households, paying the interest rate r plus the depreciation δ_k on each unit. They also rent effective labor hours N from households at wage w. Both capital K and labor N are chosen to maximize flow profits, which are given by

$$\Pi = \Theta K^{\alpha} N^{1-\alpha} - wN - (r + \delta_k)K. \tag{18}$$

Government. The government issues loans to students whose parents don't pay for all of their schooling and collects the annual payments on those loans. It also makes social security payments to the households in the economy. We assume that the government balances its budget in each period. Consequently, it levies taxes on labor income at rate $\mathcal{T}_\ell$

in order to fulfill that objective. This leads us to the budget constraint

$$\underbrace{\mathcal{T}_\ell\, wN}_{\text{Income tax receipts}} = \underbrace{\left(\sum_\omega P(\,j=c,\omega)(1-e(\omega))\right)(PT_c - p_{sl}T_{sl})}_{\text{Net student loan disbursements}} + \underbrace{(T_\ell - T_w)\left(P(\,j=c)\int \mathcal{S}(wh)dD_{c,T_{w,c}}(h) + P(\,j=n)\int \mathcal{S}(wh)dD_{n,T_{w,n}}(h)\right)}_{\text{Social security payments}}, \tag{19}$$

where $D_{j,T_{w,j}}(h)$ is the measure over human capital, conditional on education level j, in the $T_{w,j}$-th year after entering the labor force.[27] $P(\,j\,)$ is the probability of education status j in the population.

Market-clearing conditions. The market-clearing conditions for capital and labor are

$$N = \sum_\omega \sum_z \sum_{j\in\{n,c\}} P(\,j,z,\omega)\left(\sum_{t=0}^{T_w - T_c\times\mathbb{I}(j=c)-1}\int h(1-i_{j,t}(a,h,\epsilon;z,\omega))\exp(\epsilon)dD_{j,t}(a,h,e\mid z,\omega)\right) \tag{20}$$

$$K = \sum_\omega \sum_z \sum_{j\in\{n,c\}} P(\,j,z,\omega)\left(\sum_{t=0}^{T_\ell - T_c\times\mathbb{I}(j=c)-1}\int a\,dD_{j,t}(a,h,\epsilon\mid z,\omega)\right). \tag{21}$$

In the above, $i_{j,s}(a,h,\epsilon;z,\omega)$ is the time allocated by an agent with education level j to learning s periods after entering the labor force, while having idiosyncratic state (a,h,ϵ) and type (z,ω). Then $D_{j,s}(a,h,\epsilon;z,\omega)$ is the measure over idiosyncratic states for such an agent and $P(j,z,\omega)$ is the joint distribution of education levels and types.

We end with the condition for the transfer market, which is that there are zero net transfers in the economy, so that, in each year, the total amount of transfers received by children is equal to the total amount of transfers given by parents. This can be represented

[27] We multiply by the number of retirement periods because social security payments are constant throughout retirement.

by

$$0 = \sum_{\omega} \sum_{z} \sum_{j \in \{n,c\}} P(j,z,\omega) \Bigg[\mathbb{I}(j=c) \left(e(\omega) P T_c + (1 - e(\omega)) p_{sl} T_{sl} \right) + \sum_{t=0}^{T_w - T_c \times \mathbb{I}(j=c) - 1} \left(\tau_{j,t}(\omega) + w \int h \, f_{j,t}(\epsilon, \omega)(a,h,\epsilon;z,\omega)) \exp(\epsilon) dD_{j,t}(a,h,\epsilon \mid z,\omega) \right) \Bigg]. \tag{22}$$

5 What do transfers do?

In this section, we first detail the model calibration and assess the quality of the fit. We then evaluate the aggregate and distributional impacts of inter vivos transfers by simulating a counterfactual economy where parental transfers are eliminated entirely. Finally, we decompose the welfare gains of having wealthy parents to quantify the specific value children derive from parental insurance versus other early-life advantages.

5.1 Calibrating the model

We now detail the parameter calibration and the estimation of the processes that govern children's initial conditions $(h, z, \omega_c, \omega_n)$. We also specify how these initial states depend on the parents' starting human capital h_p and learning rate z_p.

Externally calibrated parameters. The values for the parameters which are not calibrated within the model are displayed in Table 3. The first block of parameters consists of those which are standard in business cycle models. The parameters that pertain to agents' preferences, the discount rate β, the EIS η_c, and the CRRA η_r, are chosen to be consistent with the values typically used for models calibrated to annual frequency. Then, the parameters governing the production sector in the economy, the capital share α and the depreciation rate δ_k, are chosen to match the real interest rate r and pre-tax wage w in the economy.

The next block concerns the dynamics of human capital and income risk over the life cycle. The parameters of the human capital production function that govern human capital depreciation ϕ and the decline in the ability to replenish it δ are set to match the observed decline in earnings near retirement and the decline in the fraction of time used for human capital accumulation over one's work life.[28] We set the scale γ of the human capi-

[28]The assumption that social security payments are a function of wages in the economy as well as one's

Model parameters			Prices, loan terms, and tax rates		
Parameter		Value	Quantity		Value
Business cycle			*Macro values*		
β	Discount rate	0.96	$R-1$	Real interest rate	2.0%
η_c	EIS	0.5	$\mathcal{T}_\ell$	Income tax rate	27.8%
η_r	CRRA	3.0	*Student loans*		
δ_k	Capital depr.	11.5%	$R_{sl}-1$	Interest rate	6.52%
α	Capital share	0.2	T_{sl}	Loan term	20 years
Human capital			*College*		
γ	Human capital scale	0.65	P	Annual cost	\$40,000
ϕ	Human capital depr.	0.005	θ	Consumption frac.	75%
δ	Learning ability depr.	0.95	T_c	College length	4 years
μ_ϵ	Mean of $\log(\epsilon)$	-0.01			
σ^2_ϵ	Variance of $\log(\epsilon)$	0.02			
Phases of life					
T_w	Years available to work	47			
T	Years as parent or child	30			
T_ℓ	Years alive	60			

Table 3: Baseline model parameters and inputs

Notes: Parameters calibrated to annual frequencies. Income tax rates and capital depreciation are determined within the model, given the other parameters, which are chosen directly, and the estimated stochastic processes.

tal production function to capture decreasing returns in human capital accumulation and the variance of idiosyncratic shocks σ^2_ϵ to capture income risk over the life-cycle. Both are chosen to remain close to the values used in Huggett et al. (2011). The mean μ_ϵ is set to normalize $\mathbb{E}[\exp(\epsilon)]$ to 1.

The remaining parameters govern aggregate economic variables. student loans, and the cost of education. The real interest rate is set at a standard level, while the income tax rate is chosen to balance the government's budget. The parameters for student loans reflect empirical averages for the rate paid on student loans and the time it takes to pay off one's loans. The cost of college P is chosen to match the total annual cost of college at a four-year institution in the US in 2026 from Hanson (2026).[29] The consumption fraction θ is calibrated to match the average fraction of the cost spent on room and board, under the assumption that half of tuition costs go toward consumption value for the student (i.e. consumption value of education).

level of human capital in the final year before retiring means that, absent learning ability depreciation (i.e. $\delta \geq 1$), agents would spend much more time learning on the job later in life than is observed in the data. Hence, $\delta < 1$ is crucial. We choose it to match the qualities of the average path of time allocated to human capital accumulation in Huggett et al. (2011), which starts at around 0.25-0.3 and steadily decreases to zero by retirement.

[29] These values are based on data from the National Center for Education Statistics (NCES).

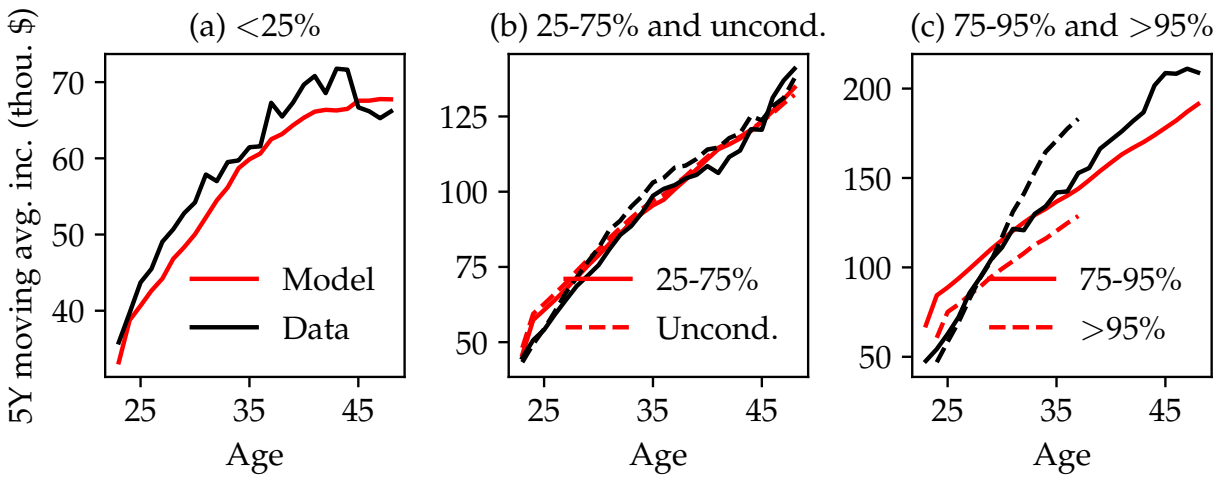


Figure 7: Five-year moving avg. income of children by parents' wealth group (model v. data)

Notes: Authors' calculations using data from the PSID. In all plots, red lines are the values in the model, while black lines are the values in the data.

Intergenerational driving processes. We assume that, given their parents' state (h_p, z_p), each child's state is drawn from the distribution $Q_{h,z}(h, z \mid h_p, z_p)$. In other words, the state (h, z) follows a first-order Markov process across generations. Given a child's state (h, z), ω_c and ω_n are drawn independently from the distributions $Q_c(\omega_c \mid h, z)$ and $Q_n(\omega_n \mid h, z)$.[30]

For a given j, $\omega_j = \{\omega_{Ins,j}, \omega_{Unc,j}, \omega_{Giving,j}\}$ is a tuple of indices. $\omega_{Ins,j}$ is the index referring to a specific set of parameters for the parental insurance the child receives over their life, while $\omega_{Unc,j}$ indexes the sequences of educational and unconditional transfers received and $\omega_{Giving,j}$ indexes the sequence of transfers that the child will give once they become a parent themselves.

It is important to note that we only estimate the transition probabilities of these processes. The values of (h, z) in the chain or the sequences indexed by each ω are predetermined. We choose them to be sufficiently rich to allow the model to match the targeted empirical moments. Additional details on these processes are provided in Appendix C.1.

The moments and the fit. We aim to understand the relationship between transfers, income, and wealth, as well as the intergenerational persistence in those quantities. Therefore, we find it natural to set the parameters of $Q_{h,z}$, Q_c, and Q_n based on conditional

[30] While Q_c and Q_n do not depend directly on the parents' state, the unconditional distributions of ω_c and ω_n remain linked to parental characteristics through the intergenerational transmission of (h, z).

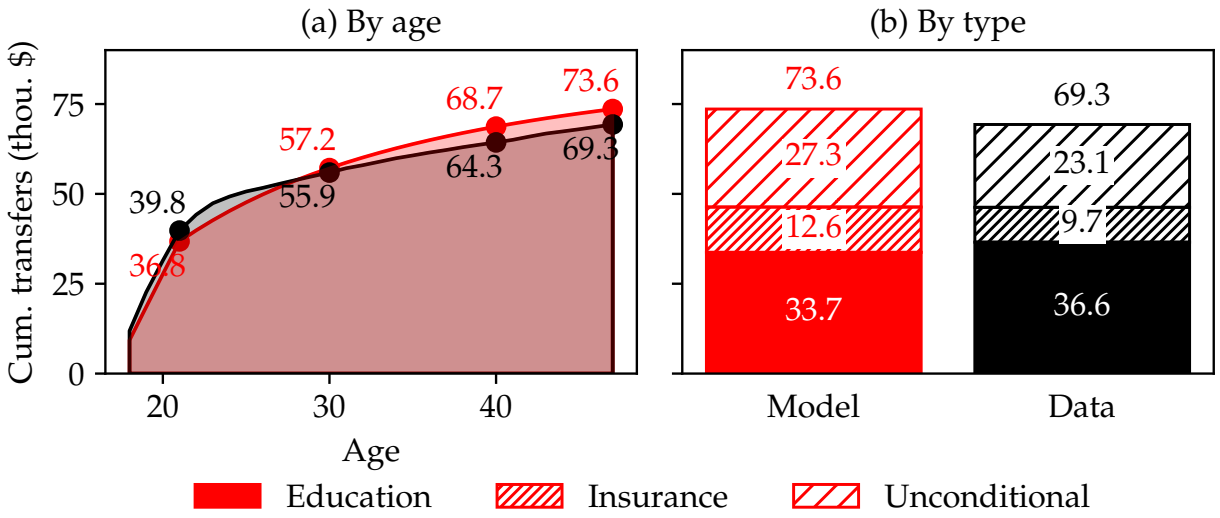


Figure 8: Average cumulative transfers received by children, by age and type (model v. data)

Notes: Authors' calculations used data from the HRS.

moments for income and transfers.

Specifically, we target the life-cycle profiles of the five-year moving average of children's income, both unconditionally and conditional on their parents' wealth group. We target analogous profiles for educational, insurance, and unconditional transfers. The parameters are jointly estimated to minimize the total root mean squared error (RMSE) across all targeted moments.

We present results showing the quality of the fit for the income moments in Figure 7 and the fit for the average transfers in Figure 8. The plots displaying the fit quality for the transfers children receive, conditional on parents' wealth group, are in Figure C.1 of Appendix C.3, along with a number of untargeted moments produced by the model.

Overall, the quality of the fit is quite good across all moments. The model is able to replicate the dependence of children's incomes on parents' wealth. It is also able to match the time path and mix of transfers received by children. Moreover, it can replicate the dependence of these values on parents' wealth groups. This tight fit gives us confidence that the exogenous processes we have specified are able to capture the most salient features of the data.

5.2 A world without transfers

We now simulate a counterfactual economy where parents no longer provide inter vivos transfers to their adult children. Mechanically, we eliminate the financial transfer processes Q_c and Q_n, leaving the transmission of initial human capital and learning ability, $Q_{h,z}$, as the sole intergenerational linkage.

By shutting down these financial channels, we can evaluate the resulting general equilibrium effects on macroeconomic prices and quantities. Furthermore, isolating $Q_{h,z}$ allows us to quantify how much the persistence of income and wealth across generations is driven by financial transfers rather than the direct transmission of human capital and learning ability.

There are two scenarios we consider in this experiment. The first and simplest is the partial equilibrium (PE) scenario, where prices and tax rates do not adjust. This is a pure decomposition exercise, measuring the contribution of parents' transfers to the determination of individual and economy-wide outcomes. Moreover, this PE environment allows us to identify the types of individuals that rely most heavily on inter vivos transfers under the status quo.

The second scenario is the general equilibrium (GE) counterfactual. In this case, we explore the effectiveness of price changes in mitigating the impacts of eliminating inter vivos transfers. In particular, we allow wages and tax rates to adjust, but we hold the Markov transition matrix $Q_{h,z}$ constant. By comparing the GE and PE outcomes, we can evaluate how endogenous price adjustments dampen or amplify the direct effects of eliminating inter vivos transfers.

How much can transfers explain? We begin by studying the importance of transfers for the levels of economic variables under each of the two scenarios. Table 4 displays the moments of the post-tax permanent income and retirement wealth distributions across the baseline, PE, and GE scenarios.

Focusing first on the PE counterfactual economy, average permanent income falls by \$326,000 relative to the baseline. This excludes the average present value of the transfers themselves, which means that the indirect effects associated with giving and receiving inter vivos transfers generate about 11% of the average post-tax permanent income earned by individuals in the status quo. Moreover, the standard deviation of permanent income is reduced by \$149,000. This corresponds to a roughly 17% decline the variance of permanent income, which we can interpret as the fraction of the variance in permanent income explained by inter vivos transfers.

Similarly, the average level of wealth at retirement falls by \$84,000 and its standard

	Baseline		No transfers (PE)		No transfers (GE)	
Moment	Perm. inc.	Ret. wealth	Perm. inc.	Ret. wealth	Perm. inc.	Ret. wealth
Mean	\$2,897k	\$626k	\$2,571k	\$542k	\$2,874k	\$580k
Std. dev.	\$1,665k	\$394k	\$1,516k	\$336k	\$1,199k	\$250k
Gini	0.32	0.35	0.33	0.34	0.23	0.24
Percentiles						
25%	\$1,211k	\$293k	\$1,033k	\$202k	\$1,420k	\$285k
Median	\$2,635k	\$512k	\$2,516k	\$493k	\$3,342k	\$647k
75%	\$3,822k	\$839k	\$3,525k	\$712k	\$3,727k	\$732k
95%	\$5,781k	\$1,367k	\$5,603k	\$1,240k	\$4,803k	\$923k

Table 4: Model-generated moments of post-tax permanent income and retirement wealth (age 65) distribution under baseline and counterfactuals

Notes: "Baseline" shows the moments generated by the estimated model, while "No transfers (PE)" and "No transfers (GE)" contain the moments when we shut down transfers, without and with price changes, respectively.

deviation drops by \$58,000. For wealth, these imply that the indirect effects of inter vivos transfers can account for 13% of the average level of retirement wealth and 26% of its variance. These numbers are striking as they imply that the indirect effects of transfers are even larger than their direct effects. The average PE effect on permanent income is almost five times the size of the average present value of transfers, about \$69,000, as shown in Table 2.

For both variables, the effects are occur disproportionately at the bottom of the distribution, though transfers also have sizable impacts on the individuals in the upper tail. Because the effecs span the entire distribution, measures of inequality in the model, like the Gini index, remain largely unaffected by the elimination of transfers in PE.

How much can prices change things? We now turn to the general equilibrium counterfactual, displayed in the rightmost columns of Table 4, to evaluate how price responses alter these outcomes. We find that, because of the large declines in human capital accumulation along the intensive and extensive margins, the post-tax wage rate rises over 30%, erasing almost all of the partial-equilibrium decline in permanent income. The overall average permanent income only falls by about \$23,000. Yet, because the aggregate saving rate remains low relative to the baseline, output is significantly depressed, by even more than is indicated by the decline in permanent income. Ultimately, price changes serve as an imperfect substitute for parental transfers in sustaining macroeconomic activity.

Furthermore, this general equilibrium wage spike precipitates a substantial decline

	Permanent income	Retirement wealth
Non-transfer effects	0.654 (91.2%)	0.531 (86.7%)
+ Insurance effects	0.031 (4.3%)	0.036 (5.9%)
+ Other transfer effects	0.033 (4.5%)	0.046 (7.4%)
= Baseline	0.717 (100%)	0.613 (100%)

Table 5: Decomposing intergenerational elasticities in the model

Notes: Fraction of total IGE accounted for in parentheses.

in both income and wealth inequality. Because households at the bottom of the wealth distribution generally have human capital stocks that are more or less fixed over the life cycle, unlike those of individuals at the top, this uniform increase in the wage rate disproportionately boosts their total post-tax labor income, compressing the overall income distribution.[31]

The role of transfers in persistence. We now analyze how financial transfers affect the intergenerational persistence of income and wealth. We quantify this persistence using the intergenerational elasticity (IGE), which measures the percentage change in a child's outcome associated with a one-percent change in their parent's outcome.

In the model, we estimate the IGEs for permanent income and retirement wealth with regressions of the form

$$\log(x_{c,i}) = \alpha + \beta \log(x_{p,i}) + \epsilon_i, \tag{23}$$

where i indexes a specific parent-child pair. The term $x_{c,i}$ denotes the outcome variable (either permanent income or retirement wealth) for the child in pair i, and $x_{p,i}$ denotes the corresponding value for their parent. The results for this exercise are collected in Table 5. In the baseline economy, the IGE of permanent income is 0.72, meaning that a 1% increase in the parent's permanent income leads to a 0.72% increase in the child's permanent income. The baseline IGE for retirement wealth is 0.61

Our baseline IGE of permanent income (0.72) closely aligns with the empirical estimate of 0.66 established by Mazumder (2016). Because this estimate is constructed using long, 15-year averages of mid-career income for both parents and children, it serves as a reasonable empirical proxy for our model's concept of permanent income. While some

[31] This uniform wage increase relies on the assumption of perfect substitutability between college and non-college labor. Introducing imperfect substitutability would likely widen the college wage premium in response to the reduced supply of educated labor, which could counteract this progressive redistributive effect.

structural models (e.g., Lee and Seshadri, 2019) target lower IGE estimates such as the 0.34 found by Chetty et al. (2014), those values rely on short-run snapshots of early-career income. As discussed by Mazumder (2016) and Mogstad and Torsvik (2023), short-run income measures suffer from significant downward life-cycle bias when used to approximate lifetime resources, making them less appropriate benchmarks for our permanent income elasticities.

Table 5 decomposes the IGE of permanent income and retirement wealth into the fraction driven by inter vivos transfers versus all other factors. We further isolate the specific contribution of the insurance effect of these transfers. We find that transfers account for almost 9% of the IGE of permanent income and 13% of the IGE for wealth at retirement. The insurance effect accounts for for 49% of the transfer effect for permanent income and 44% for wealth. Overall, in both cases, transfers account for a considerable fraction of the IGE and about half of that effect comes from their insurance effects.

In the previous exercise we assessed the intergenerational persistence of absolute income and wealth levels. Now, we consider cross-generational transition matrices, which provide an account of the level of wealth and income mobility in the economy by assessing the dependence of children's economic ranks on the ranks of their parents. In Tables 6 and 7, under the 'Baseline' heading, we present the matrices generated by the model for retirement wealth and permanent income.

Under the 'No transfers (PE)' heading, we perform an experiment to understand transfers' contributions to children's ability to obtain particular ranks in the distributions. Supposing that the distribution of initial human capital and learning rates (h, z) for parents, given their wealth rank, is the same as in the 'Baseline' scenario, we ask: what is the probability that the child of a parent in a particular wealth group makes it into each wealth group if that child never receives any inter vivos transfers? How do those probabilities depend on parents' wealth groups? Across wealth groups, we find that transfers are crucial for intergenerational mobility. However, Table 6 indicates that they appear far more central to the maintenance of rank at the tops of the distributions. For children whose parents are in the top 5% of the wealth distribution, removing transfers reduces the chance of staying in the top wealth quartile by over 20 percentage points. The same is true for the children of parents in the 75-95th percentiles.

When we perform the same exercise for permanent income, categorizing parents on the basis of their permanent income group, the results are qualitatively the same, though the magnitudes of the effects are even larger. This is unsurprising when we consider that, by retirement, wealthier parents happen also to have given a substantial amount of resources to their children. This pushes against both the indirect effects, which raise perma-

		Baseline				No transfers (PE)			
	Child grp.	<25%	25-75%	75-95%	>95%	<25%	25-75%	75-95%	>95%
Parent grp.	<25%	0.51	0.49	0.0	0.0	**0.59**	**0.41**	**0.0**	**0.0**
	25-75%	0.24	0.48	0.21	0.07	**0.32**	**0.5**	**0.17**	**0.01**
	75-95%	0.0	0.5	0.4	0.09	**0.0**	**0.74**	**0.24**	**0.02**
	>95%	0.0	0.64	0.33	0.03	**0.0**	**0.88**	**0.12**	**0.01**

Table 6: Probability of child retirement wealth group, given retirement wealth group, baseline v. partial eq. counterfactual

Notes: Retirement wealth is the level of assets held at age 65, when agents retire. Bold entries in the transition matrix for the counterfactual represent values that change relative to the baseline. Those in black have increased. Those in red have decreased.

		Baseline				No transfers (PE)			
	Child grp.	<25%	25-75%	75-95%	>95%	<25%	25-75%	75-95%	>95%
Parent grp.	<25%	0.53	0.47	0.0	0.0	**0.61**	**0.39**	**0.0**	**0.0**
	25-75%	0.22	0.48	0.22	0.07	**0.3**	**0.52**	**0.17**	**0.01**
	75-95%	0.0	0.5	0.41	0.09	**0.0**	**0.77**	**0.21**	**0.02**
	>95%	0.0	0.66	0.31	0.03	**0.0**	**0.93**	**0.07**	**0.01**

Table 7: Probability of child permanent income group, given parent's permanent income group, baseline v. partial eq. counterfactual

Notes: Bold entries in the transition matrix for the counterfactual represent values that change relative to the baseline. Those in black have increased. Those in red have decreased.

nent income, as well the mechanical increase in wealth that comes from receiving more transfers. Because of the presence of all these competing forces on wealth, we would expect permanent labor income ranks in the model, which are only indirectly affected by transfers, to have a stronger correlation than we observe for wealth ranks.

Across all of these exercises, the results are remarkably consistent: inter vivos transfers have strong indirect effects on the wealth and permanent income of their recipients, effects that are about four to five times the present value of the transfers themselves. They also contribute meaningfully to the intergenerational persistence of both absolute levels and relative ranks, particularly at the upper tail of the distribution.

5.3 The welfare value of transfers

Having quantified the role of inter vivos transfers in shaping aggregate outcomes and intergenerational persistence, we now turn to their welfare implications for the recipients. How valuable are these transfers relative to other sources of variation? Furthermore,

which specific transfer channels generate the greatest welfare gains for children? We answer these questions by evaluating the welfare equivalence of transfers and isolating the margins that provide the highest marginal value to recipients

The value of having wealthy parents. The value function we obtain in solving the model is in terms of a consumption equivalent $W_j(h, z, \omega)$. So, the value of having parents in a particular wealth bracket q_p is determined by these values and the distribution[32]

$$P(h, z, \omega, j \mid q_p) = P(\omega, j \mid h, z)\, P(h, z \mid q_p). \tag{24}$$

That is, having wealthy parents influences the distribution of human capital h and the learning rate z, which also, in turn, influences the distribution over transfer types ω and education levels j.

With all of these channels for influence, there is a natural point of comparison. We imagine a child drawing their endowment of (h, z) from the unconditional distribution $P(h, z)$ in a world where they won't receive inter vivos transfers from their parents. For each q_p, how much would their value be changed by drawing their endowment from $P(h, z \mid q_p)$ in a world with transfers?

Figure 9 shows the results from this experiment in the red dots in subplot (a). Regardless of with wealth group you select into, transfers mean you're still better off. More concretely, drawing from $P(h, z \mid q_p)$ is worse than drawing from $P(h, z)$ for all but the two highest wealth groups $q_p = \{$75-95%, >95%$\}$, but the potential to receive transfers can make up for the difference.

Figure 9 also displays the effects decomposed[33] into the parts that can be accounted for by differences in the distribution of initial human capital h by parent group, in the distribution of learning rates z, transfers, and college attendance rates.

We observe that variation in the distribution of learning rates, which increase the returns to college and the growth rates of income while in the work force, is the main driver behind the value of having parents in a higher wealth group. The value from transfers makes up a smaller, but still major, portion of the increase in value. Variation in the distribution of initial human capital and changes in the probability of college attendance account for even smaller portions of the change in value.

Interestingly, the increase in value from receiving transfers is relatively uniform across parents' wealth groups, even though children of parents in higher wealth groups receive

[32] The decomposition follows from Bayes' rule and the fact that $P(\omega, j \mid h, z, q_p) = P(\omega, j \mid h, z)$

[33] We follow Shorrocks et al. (2013), Lundberg and Lee (2017), and Bhandari et al. (2026) in using Shapley values to compute our decompositions.

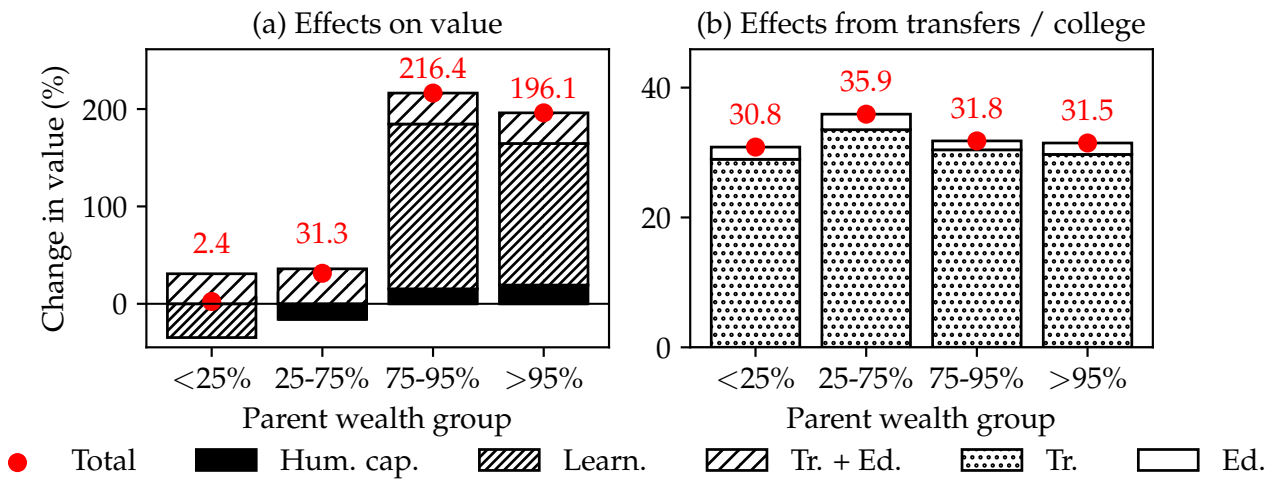


Figure 9: Decomposing the value of having wealthy parents

Notes: Subplot (a) combines transfer effects (Tr.) and education effects (Ed.) into one. Subplot (b) shows the split between transfers and education effects. Scale is relative to unconditional baseline without transfers.

much more in transfers than children of parents in lower wealth groups. The much higher marginal value of transfers for the children of poorer parents almost exactly balances out the lower level of transfers received, so that inter vivos transfers generate large increases in value across the wealth distribution.

The value of insurance transfers. In this paper, we have emphasized the importance of parental insurance, but who do insurance transfers benefit the most? We evaluate this in terms of the average increase in value generated by having the most comprehensive insurance transfer type relative to a life without transfers, shown in Figure 10(a).

The marginal value of insurance is generally increasing in parental wealth group for those who don't attend college. That pattern is a natural implication of the fact that overall early income risk is higher for those with higher initial levels of human capital. For those who do attend college, the relationship is flipped. This is still intuitive. Given that they attend college, student loan payments make up a large fraction of the income earned by children of the lower wealth groups. For these individuals, who very often find themselves constrained, negative shocks are especially painful, but generous parental insurance can go a long way in softening the blows.

Though the benefits of insurance are strongest for the poorest, the logic behind its disproportionately large value holds for the vast majority of parental wealth groups. For all but college attendees whose parents are in the 75-95% group, insurance transfers generate increases far in excess of their size relative to labor income, which is uniformly around

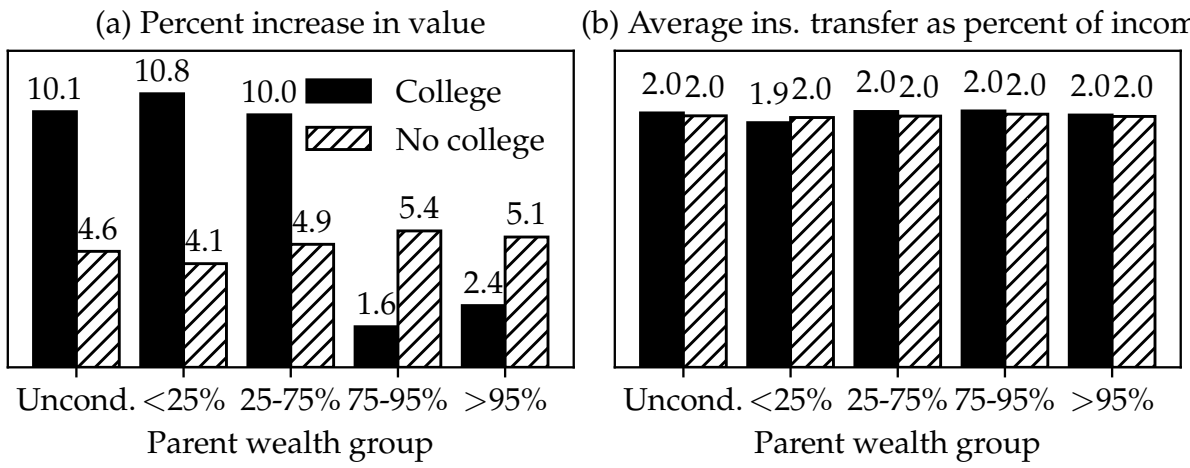


Figure 10: The value of generous parental insurance

Notes: Subplot (a) shows how the value increases, relative to a life without any transfers. Subplot (b) shows the average insurance transfer as a percent of labor income in each period.

2%. Of course, their value is lower for those whose lifetime incomes are expected to be especially high and who can insure themselves to a reasonable degree, but they remain a powerful source of insurance for the vast majority of individuals.

6 Conclusion

In this paper, we argue that inter vivos transfers (IVTs) from parents to children play a substantially larger role in driving the cross-sectional dispersion and intergenerational persistence of income and wealth than their raw magnitudes imply. Using survey data, we document that the disproportionate impact of IVTs stems from two distinct features. First, they are heavily front-loaded toward the beginning of the recipient's adult life, generating a mechanical present-value advantage relative to other types of transfers. Furthermore, a large proportion of these early transfers go toward human capital accumulation, amplifying their effects through increases to lifetime income. Second, IVTs serve as an informal form of insurance. Wealthy parents, in particular, increase transfers when their children experience adverse labor market events or other financial shocks. We provide empirical evidence that the presence of this parental safety net fundamentally alters children's risk-taking behavior, incentivizing them to allocate a larger share of their portfolios to equities.

To quantify these mechanisms, we embed them in a structural overlapping generations model with incomplete markets and endogenous human capital accumulation. The model matches both the unconditional and conditional income and transfer paths observed in

the data well. Counterfactual exercises reveal that eliminating IVTs yields declines in lifetime income across the population, driven by reductions in college attendance and intensive human capital investments. The ability of parents to front-load transfers and insure against downside risk is critical in shielding children from binding liquidity constraints during peak investment years. Furthermore, we find that the implications for intergenerational wealth mobility are highly heterogeneous. While shutting down IVTs generally increases wealth mobility for the middle and upper classes by severing a primary mechanism of wealth transmission, it decreases mobility for the poor. Without parental transfers, children from the poorest households face severe difficulties in self-financing educational investments, effectively trapping them at the bottom of the distribution.

Ultimately, our findings suggest that the timing and state-contingency of intergenerational transfers are just as vital as their overall size. Recognizing IVTs as both a funding source for early-life investments and a substitute for self-insurance offers a clearer lens through which to evaluate the persistence of inequality. This framework highlights the need to carefully consider policies that alter the timing of wealth transmission—such as the taxation of gifts versus estates or the treatment of stepped-up basis—as these policies potentially have downstream effects on human capital formation, aggregate productivity, and intergenerational mobility.

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A Appendix to section 2

A.1 Bequests.

We estimate lifetime bequests from the recipient child's perspective using data from the Health and Retirement Study (HRS) exit interviews. Because these exit interviews rely on proxy respondents—typically surviving family members who may lack complete or precise financial information—they systematically underreport or completely miss a portion of actual inheritances. To correct for this attrition and reporting gap, we calculate the average observed transfer (explicitly including verified zero-dollar transfers) by parent age and wealth group, and reweight these averages using the parent's unconditional probability of death at each age derived from Social Security life tables. The total household bequest is constructed by combining the deceased parent's net estate value and any life insurance settlements. To calculate the net estate, we first adjust the reported gross estate to include primary and secondary home values if they were excluded from the raw estate variable. Next, we subtract funeral and death expenses to determine the gross taxable base. We estimate federal estate tax liabilities by applying the historical exemption thresholds and marginal rates corresponding to the year of death, and we deduct this estimated liability from the estate. Life insurance payouts are excluded from this tax calculation because they pass directly to beneficiaries outside of the probate estate. Ultimately, the total after-tax household bequest is split equally among all of the donor's children, and each individual child's share is attributed to their specific age at the time of the parent's death.

To construct the cumulative lifetime bequest profiles overall and across parental wealth groups, we map the expected household bequests onto the child's life cycle. Because the subset of observed exit interviews is too small to reliably estimate average transfers for every individual year of death, we pool the observed bequests into two broad parent death-age bins: deaths occurring at age 80 or younger, and deaths occurring at age 81 or older. We compute these averages separately for first and second household deaths, as well as conditionally by parental wealth group.

To recover a smooth, age-specific distribution of expected inheritances, we multiply these binned averages by the exact, age-specific mortality probabilities derived from the Social Security life tables. Finally, each child receives an equal share of their parent's expected bequest at each possible death age, which is then mapped to the child's own chronological age using the baseline parent-child age gap. Summing these age-specific expected values yields the final cumulative lifetime bequest curves. Figures A.1 and A.2 plot the cumulative average lifetime bequests for all children, and broken out by parental wealth group, respectively.

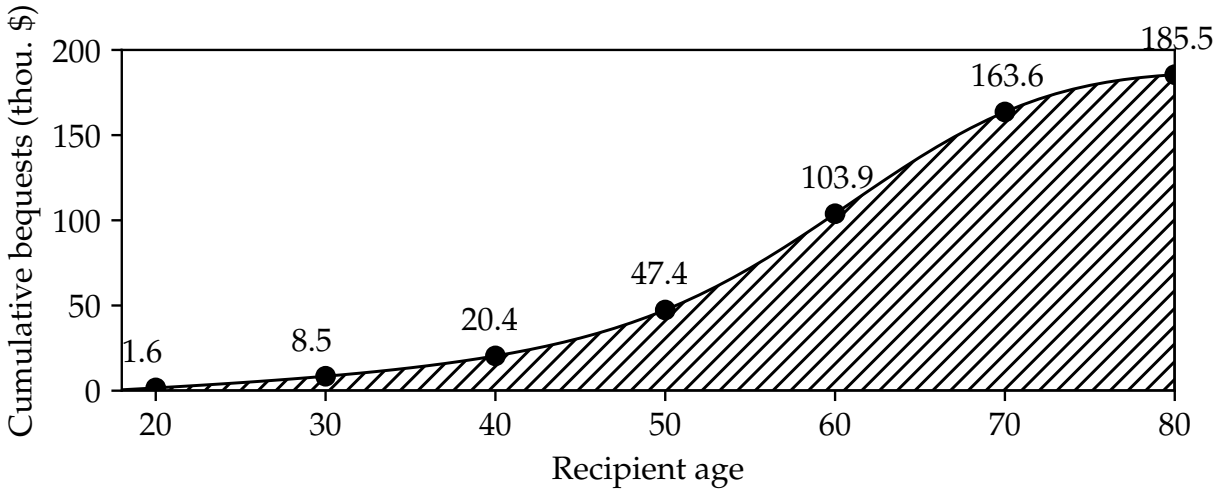


Figure A.1: Average life cycle bequests

Notes: This figure plots the cumulative average lifetime bequests by recipient (child) age.

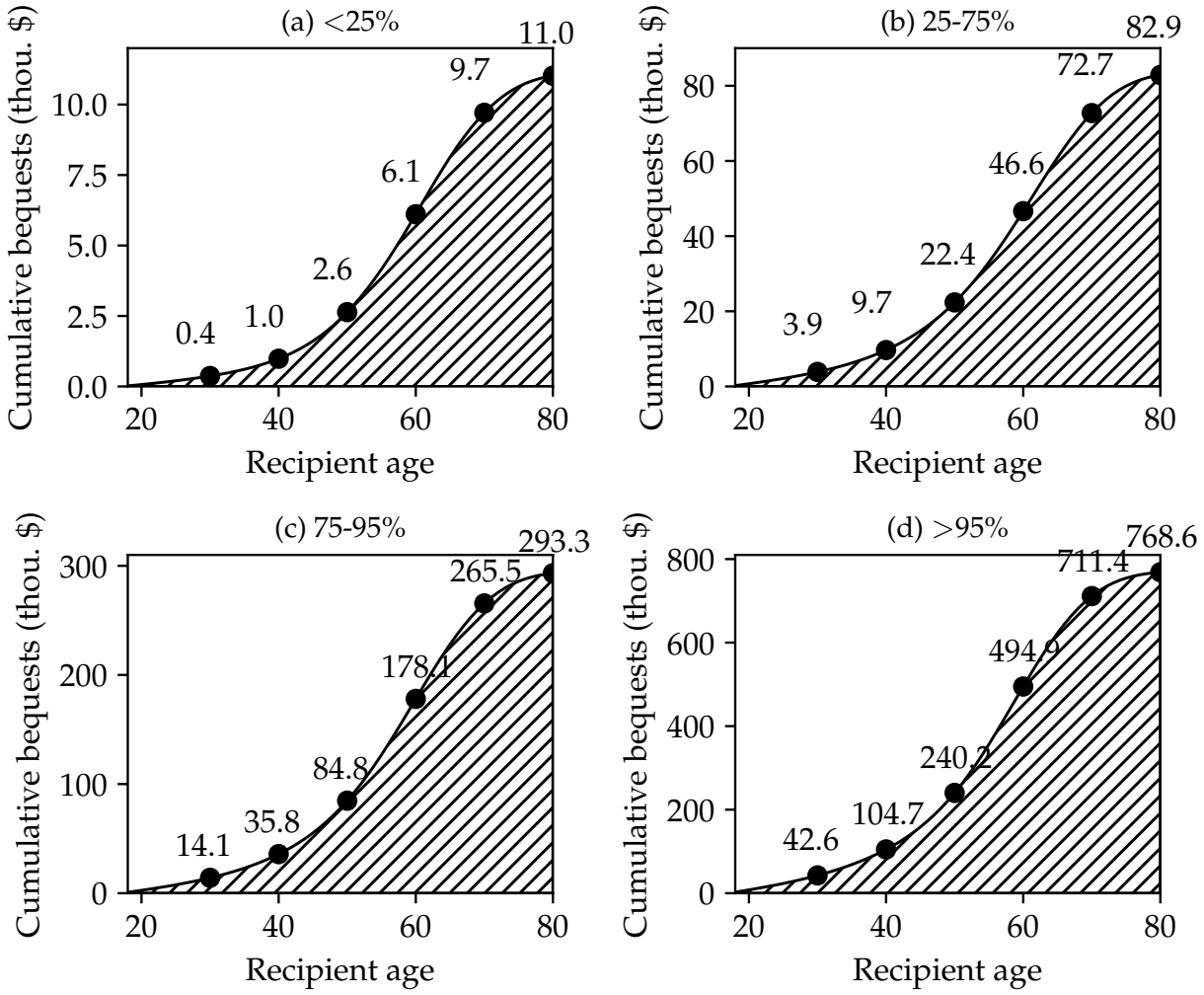


Figure A.2: Average life cycle bequests by parent wealth group

Notes: This figure plots the cumulative average lifetime bequests by recipient (child) age and donor (parent) wealth group.

	(1) Avg	(2) 0-25th	(3) 25-75th	(4) 75-95th	(5) >95th
Job Loss	65.656	-25.080	79.562	201.362	-852.775*
	(63.838)	(22.604)	(57.260)	(222.189)	(437.017)
Work Down	104.767	-40.634*	48.486	-12.271	-83.536
	(80.520)	(24.250)	(97.795)	(184.799)	(600.347)
New Child	424.124***	85.421*	404.584***	51.743	3,268.321**
	(100.447)	(49.814)	(98.534)	(264.276)	(1,594.894)
Divorce	-157.645*	-25.132	-167.443**	-419.514	550.073
	(95.554)	(52.956)	(80.713)	(267.889)	(931.380)
Marriage	36.302	10.289	49.348	-80.564	-658.446
	(71.793)	(25.519)	(66.901)	(211.608)	(639.420)
Age 18-30	1,284.089***	367.035***	854.669***	1,664.015***	4,388.585***
	(82.219)	(45.341)	(73.747)	(167.145)	(832.017)
Age 31-54	364.038***	58.915**	214.825***	347.409***	2,764.158***
	(56.750)	(23.575)	(29.537)	(76.835)	(872.383)
Age 18-30 X Job Loss	1,194.754***	-24.037	573.429**	2,217.685***	6,960.479***
	(301.042)	(85.388)	(270.811)	(779.513)	(2,513.244)
Age 31-54 X Job Loss	328.557***	180.642***	92.734	682.968*	2,654.371**
	(105.225)	(45.873)	(73.622)	(353.261)	(1,233.551)
Age 18-30 X Work Down	1,366.676*	860.652**	472.787**	538.300	10,145.838
	(745.806)	(424.271)	(216.345)	(574.855)	(7,876.220)
Age 31-54 X Work Down	187.650	150.460***	147.489	517.543*	197.412
	(120.037)	(54.066)	(125.730)	(292.058)	(1,025.590)
Age 18-30 X New Child	-1,087.171***	-107.962	-750.477***	-1,066.734**	-4,495.235
	(185.229)	(155.158)	(150.214)	(444.320)	(2,767.821)
Age 31-54 X New Child	-194.442	-3.448	-304.385***	211.900	-2,575.911
	(120.511)	(49.816)	(101.473)	(292.896)	(1,971.764)
Age 18-30 X Divorce	-126.187	-223.016**	13.651	-612.917	15,502.454
	(591.723)	(107.749)	(207.196)	(569.876)	(17,791.825)
Age 31-54 X Divorce	171.226	-57.286	173.342*	922.971***	356.633
	(117.865)	(53.749)	(96.531)	(345.087)	(1,386.681)
Age 18-30 X Marriage	532.321**	-69.939	385.706**	761.634*	1,550.078
	(216.405)	(112.795)	(191.061)	(454.666)	(1,845.707)
Age 31-54 X Marriage	397.127***	85.274*	214.223**	812.993**	2,329.881
	(126.351)	(49.428)	(87.748)	(339.808)	(1,672.536)
Constant	492.881***	64.525***	271.504***	1,086.870***	1,646.010***
	(26.198)	(20.196)	(21.567)	(62.211)	(253.157)
Observations	205,098	44,269	108,000	42,563	10,266
R-squared	0.012	0.018	0.017	0.016	0.016
Demographic FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes

Table A.1: Life events and transfers by wealth

Notes: This table reports the coefficients and standard errors from estimating equation 2 for a large set of life events. The dependent variable is total transfer amount (2024 USD). Job Loss, Work Down, New Child, Divorce, and Marriage are indicator variables equal to 1 when the recipient experienced each event between consecutive surveys.

Life events and transfer robustness.

	(1) Aggregate	(2) 0-25th	(3) 25-75th	(4) 75-95th	(5) >95th
Severe Inc Loss	657.956***	199.714*	625.180***	897.215**	2,012.645**
	(147.912)	(103.630)	(242.229)	(422.185)	(813.532)
New Child	79.203	261.393*	-98.004	213.531	573.184
	(102.051)	(148.389)	(89.836)	(242.193)	(811.984)
Divorce	454.280**	281.425	495.941*	594.854	991.940
	(204.294)	(183.214)	(270.092)	(1,201.186)	(1,498.735)
Marriage	132.352	-211.826	225.512	211.799	503.763
	(130.174)	(137.031)	(159.784)	(398.286)	(714.996)
Constant	496.186***	244.406***	428.933***	636.397***	1,281.233***
	(31.224)	(29.777)	(42.094)	(88.012)	(168.102)
Observations	18,582	6,086	9,328	1,948	1,220
R-squared	0.015	0.024	0.019	0.017	0.120
Demographic FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes

Table A.2: Life events and transfers by wealth, PSID

Notes: This table reports the coefficients and standard errors from estimating equation 2 for a large set of life events, while excluding age group indicators (due to lack of observations of older recipients). The dependent variable is total transfer amount (2024 USD). Severe Inc Loss is an indicator variable equal to 1 when the recipient's income dropped by more than 50% between consecutive surveys. New Child, Divorce, and Marriage are indicator variables equal to 1 when the recipient experienced each event between consecutive surveys.

B Appendix to section 3

B.1 Sectoral income profiles and uncertainty measurement

Sector classification and code harmonization. Following the general methodology of Boar (2021), we categorize household heads into 22 distinct employment sectors by interacting major industry and occupation classifications. Because the PSID utilizes multiple occupational and industry coding schemes across its survey waves, we harmonize the data across three distinct eras. Specifically, survey years prior to 2002 use 1970 Census classifications, survey years between 2002 and 2016 use 2000 Census classifications, and survey years after 2016 use the 2018 occupation and 2017 industry Census classifications. To link these definitions, we use the crosswalk codes provided by the Census. While our underlying coding scheme identifies unemployed individuals, we drop these observations from our main analysis to focus strictly on active occupational risk.

To construct the sectors, we first allocate workers into eight major industry groups: agriculture and mining; construction; manufacturing; transportation and utilities; trade; finance; services; and public administration. We simultaneously classify workers into six major occupation groups: management, business, and financial operations; professional specialty; technicians and administrative support; sales and services; production operators, fabricators, and laborers; and farming, forestry, and fishing.

We refine these broad baseline categories by introducing several detailed splits to better capture relevant variation in income risk. Within the manufacturing industry, we separate durable from nondurable goods. Within the services industry, we partition professional specialty occupations into three distinct sectors: computer, engineering, science, and legal professions; education, community, and arts professions; and health diagnosing and treating practitioners. Finally, to accurately capture entrepreneurial risk, we apply a categorical override to identify business owners, whom we further disaggregate into four distinct self-employment sectors based on their underlying industry: agriculture, mining, and construction; goods and distribution; business, personal, and entertainment services; and finance and professional services. Any household head reporting self-employment on their main job is assigned to one of these four dedicated sectors, superseding their standard occupation assignment. Table B.1 summarizes the industry, occupations splits.

Table B.2 reports the percentage of household heads in the PSID sample from 1968 to 2019 that reported working their main job in each sector. Statistics are shown for the aggregate population and broken out by parental wealth group. Those with wealthier parents tend to work in management, finance, professional specialties, and entrepreneurial ventures. For example, individuals with parents in the top 5% of the wealth distribution

Industry	Mgmt. & Bus.	Professional Specialty			Tech. & Admin.	Sales & Services	Prod. & Laborers	Farming & Fish.	Self-Employed
		STEM/Law	Educ./Arts	Health					
Agriculture & Mining	Sector 1							Sector 1	Sector 22
Construction	Sector 2						Sector 3	Sector 1	Sector 22
Manufacturing, Nondur.	Sector 4				Sector 6	Sector 4	Sector 6	Sector 1	Sector 23
Manufacturing, Durable	Sector 5				Sector 7	Sector 5	Sector 7	Sector 1	Sector 23
Transport & Utilities	Sector 8					Sector 9		Sector 1	Sector 23
Trade	Sector 10				Sector 11	Sector 12	Sector 11	Sector 1	Sector 23
Finance	Sector 13							Sector 1	Sector 25
Services	Sector 14	Sector 15	Sector 16	Sector 17	Sector 18	Sector 19	Sector 20	Sector 1	Sectors 24, 25
Public Administration	Sector 21							Sector 1	Sector 25

Table B.1: Sector classification by industry and occupation

Notes: This table outlines the classification of household heads into 25 active employment sectors based on the interaction of their industry and occupation. Any household head reporting self-employment on their main job overrides their standard occupation assignment and is instead classified into one of the four business owner sectors (Sectors 22-25) corresponding to their broad industry. Self-employed service workers are allocated to Sector 24 (business, personal, and entertainment services) or Sector 25 (professional services).

are highly concentrated in management and business services (10.06%), finance (7.29%), and professional STEM and legal fields (5.44%). This group is also more likely to be self-employed than, particularly in agriculture, mining, and construction (11.66%) or finance and professional services (5.92%). Those with less wealthy parents tend to work in production, manual labor, and lower-wage service roles. Individuals from the bottom 25% parental wealth group most frequently work in sales and services (10.09%), durable manufacturing production (7.15%), and trade technology and production (6.54%). Finally, the probability of being unemployed at the time of the survey is monotonically decreasing in parental wealth, falling from 9.87% for the lowest wealth quartile to 3.87% for the top 5%

First-stage earnings sample and measurement details. The sample used to construct the sector-level expected income and income uncertainty measures consists of household heads between ages 22 and 65 observed from 1968 through 2019. As in Boar (2021), we restrict this first-stage estimation to the original SRC core sample, explicitly excluding the Survey of Economic Opportunity (SEO) subsample, the 1990-1995 Latino sample, and the immigrant refresher samples. To accurately capture the full scope of income risk, we retain both employed and unemployed household heads. Finally, we require sufficient longitudinal data for each respondent by dropping any individual with fewer than three usable observations. Our primary earnings measure is the household head's total annual labor income. Because the PSID changed its reporting structure in later survey waves by separating basic labor income from the labor portion of business income, we harmonize these variables to ensure our measure consistently captures total labor earnings, including the labor part of business equity, across all years. We deflate this earnings measure using the CPI. To address the PSID's shift to biennial interviewing after 1997, we reconstruct a

Sector	All (%)	0-25th (%)	25-75th (%)	75-95th (%)	>95th (%)
Unemployed	6.22	9.87	5.82	3.79	3.87
Agriculture/Mining, Farming	2.04	2.16	2.14	1.62	2.22
Construction, Exec/Prof/Tech/Admin	1.43	0.62	1.73	1.33	2.45
Construction, Production/Laborers	3.82	3.91	4.18	3.57	1.04
Manufacturing, Nondurable, Mgmt/Sales	1.62	0.65	1.54	2.54	2.92
Manufacturing, Durable, Mgmt/Sales	3.07	1.45	2.74	5.31	4.17
Manufacturing, Nondurable, Tech/Prod	3.09	3.85	3.88	1.03	0.67
Manufacturing, Durable, Tech/Prod	5.50	7.15	5.94	3.59	1.92
Transport/Utilities, Exec/Prof/Tech	3.62	3.03	3.67	4.30	3.03
Transport/Utilities, Sales/Production	4.53	5.65	4.72	3.73	1.08
Trade, Exec/Prof	1.67	1.34	1.53	2.27	2.05
Trade, Tech/Admin and Production	4.64	6.54	4.72	3.16	1.61
Trade, Sales/Services	5.56	7.11	5.64	4.23	3.46
Finance	4.80	2.97	4.82	6.10	7.29
Services, Mgmt/Business/Financial	5.02	2.21	4.78	7.38	10.06
Services, Prof: Computer/Eng/Science/Legal	2.34	1.13	1.88	3.97	5.44
Services, Prof: Education/Community/Arts	5.80	4.36	5.95	6.79	6.61
Services, Prof: Health	2.18	1.01	2.40	3.15	1.29
Services, Tech/Admin	4.74	5.04	4.89	4.49	3.03
Services, Sales/Services	6.58	10.09	6.19	4.19	4.61
Services, Production/Laborers	2.23	3.02	2.10	2.23	0.11
Public Administration	6.07	5.35	6.91	5.07	5.29
Self-Employed, Agri/Mining/Construction	4.43	3.51	3.70	5.30	11.66
Self-Employed, Goods and Distribution	2.42	2.35	2.11	2.47	5.41
Self-Employed, Bus./Personal/Ent. Services	3.20	3.74	2.91	3.43	2.77
Self-Employed, Finance and Prof. Services	3.38	1.89	3.12	4.95	5.92

Table B.2: Sector composition

Notes: This table reports the percentage of household heads employed in each sector between 1968 and 2019.

continuous annual earnings panel by utilizing the retrospective income questions covering the two years prior to the interview, carefully adjusting the reporting age to align with the actual calendar year the income was earned. We winsorize raw earnings at the 1st and 99th percentiles to mitigate the influence of extreme outliers. Furthermore, because a strict lower bound of zero earnings prevents logarithmic transformations and masks the impact of severe extensive-margin unemployment shocks, we impute zero and near-zero earnings (below $200) using a demographically-projected government transfer function.

B.2 Main sample construction and financial variable definitions

Child-parent matched sample. Our main analysis sample consists of child-parent pairs drawn from the PSID. To construct the intergenerational links, we match children to their

parents using the PSID's invariant, individual-level parent identification variables. We restrict the child sample to household heads aged 18 to 65. We utilize survey waves from 1999 onward, aligning with the PSID's introduction of an expanded consumption measure. The underlying data includes both the SRC core sample and the Survey of Economic Opportunity (SEO) subsample, but we explicitly exclude the 1990 to 1995 Latino sample as well as the 1997, 1999, and 2017 immigrant refresher samples. Incorporating these later cohorts would introduce selection biases due to our intergenerational linkage mechanism. Because our empirical strategy requires observing parental wealth during the child's adolescence, individuals from these newer samples would be systematically underrepresented. Finally, to focus the analysis on active occupational risk, we require household heads to be employed in a defined sector and exclude unemployed individuals.

For our two-stage least squares (2SLS) identification strategy, we construct an initial parental wealth instrument anchored to the child's youth. Specifically, we measure parental wealth in the survey year where the child's age is closest to 16. We restrict this search to a valid age window between 10 and 22. If there is an exact tie in age distance from 16, we select the observation corresponding to the later age. This procedure prioritizes the father's household wealth, falling back to the mother's household wealth only if the father is unobserved during the entire 10 to 22 age window. Because this instrument is anchored to the specific year a given child reaches this age target, siblings born in different years are assigned parental wealth values drawn from their respective calendar years, ensuring each child's instrument reflects the financial resources available during their specific adolescence.

Asset portfolios, business equity, and savings. We construct the three primary measures of household financial risk-taking and precautionary behavior using the PSID's wealth, income, and expenditure modules. First, we define the risky asset share as the ratio of a household's risky assets to its total assets. Risky assets are strictly defined as the sum of stock holdings (including mutual funds and investment trusts) and business or farm equity. The denominator, total assets, is calculated as the sum of these risky assets alongside checking and savings accounts, money market funds, certificates of deposit, savings bonds, Treasury bills, IRAs and annuities, real estate (including primary and secondary homes), and vehicles. Second, we capture entrepreneurial risk-taking using a binary indicator that equals one if any family member owns a business or holds a financial interest in a business enterprise. This ownership measure is distinct from the self-employment classification used in the sectoral analysis, as it captures financial equity regardless of active

labor participation.

Third, we measure the household savings rate as disposable family income (total income net of family taxes) minus total family consumption, divided by disposable income. Because the PSID substantially expanded its consumption expenditure questions in 2005 to include new categories—specifically telephone and internet, home repairs, furnishings, clothing, trips, and recreation—we must impute these expenditures for the 1999, 2001, and 2003 survey waves to construct a consistent consumption basket alongside the continuously observed categories of food, housing, utilities, transportation, education, childcare, and health care. Following the methodology of Attanasio and Pistaferri (2014), we regress the log of the spending on these newer categories on a cubic polynomial of the sum of the continuously observed categories, alongside demographic controls and local price indices, using the 2005 to 2023 waves. We use the exponentiated fitted values from this model to predict the missing consumption components for the early waves, adding them to the observed consumption totals. To mitigate the influence of extreme outliers, we drop observations in the bottom 5 percent of the disposable income distribution and trim the top and bottom 2.5 percent of the resulting savings rate distribution. Finally, all monetary values are deflated to 2024 dollars using the CPI.

Table B.3 summarizes the distributions of the specific portfolio allocation, business ownership, and savings measures constructed above, reporting means and standard deviations for both the aggregate sample of children and conditional on their parents' lifetime wealth. Every margin rises with parental wealth. Children of parents in the top 5 percent hold about four times the equity share of children of parents in the bottom quartile (0.19 versus 0.04). They are more than four times as likely to hold any stock or business equity (56 versus 13 percent) and about three times as likely to own a business (26 versus 8 percent). Saving rates also rise with parental wealth, from 19 to 28 percent. Differences in levels are larger still: mean total assets are about six times higher and mean risky assets about nine times higher in the top group than in the bottom quartile, while consumption and disposable income are roughly twice as high. The dispersion of asset holdings is large in every group.

		Parent Lifetime Wealth			
	All	0-25th	25-75th	75-95th	>95th
Portfolio, business ownership and saving					
Equity share of total assets	0.083	0.044	0.072	0.123	0.192
	(0.195)	(0.150)	(0.181)	(0.227)	(0.268)
Holds equity (stock or business)	0.259	0.125	0.236	0.401	0.563
	(0.438)	(0.330)	(0.424)	(0.490)	(0.496)
Business owner	0.134	0.083	0.126	0.181	0.263
	(0.341)	(0.275)	(0.332)	(0.385)	(0.441)
Saving rate	0.243	0.193	0.251	0.269	0.279
	(0.325)	(0.342)	(0.319)	(0.315)	(0.315)
Levels (*thousands of 2024 dollars*)					
Total assets	414.0	170.9	305.0	795.9	1,089.4
	(1,637.0)	(797.3)	(738.8)	(3,083.8)	(2,056.5)
Risky assets (stock + business equity)	135.4	47.7	73.7	316.8	424.7
	(1,228.2)	(515.4)	(449.7)	(2,447.3)	(1,380.3)
Consumption	64.1	48.9	61.8	80.5	89.9
	(47.9)	(38.0)	(43.7)	(55.9)	(60.9)
Family income	117.2	78.0	111.4	163.7	173.8
	(153.8)	(102.3)	(140.7)	(206.9)	(165.7)
Disposable income	98.1	68.8	94.3	131.7	138.5
	(105.2)	(73.3)	(97.1)	(136.9)	(114.5)
Child-years	46,328	16,185	21,892	6,674	1,577

Table B.3: Summary statistics: portfolios, business ownership, and saving

Notes: This table reports the sample means and standard deviations for measures of asset allocation, business ownership, savings, and income. The sample consists of linked children (household heads aged 18 to 65 with observable lifetime parental wealth) from the 1999-2023 PSID waves. Standard deviations are reported in parentheses below the means.

C Appendix to section 5

C.1 More details on processes and transfer types

Grids for human capital and the learning rate. Our empirical results indicate that variation in learning rates is a greater source of dispersion than variation in initial human capital, so we use more points for learning rate than initial human capital. We use three grid points for initial human capital $h \in \{15, 30, 60\}$ and six for the learning rate $z \in \{0.1, 0.3, 0.5, 0.7, 1.0, 1.5\}$. This yields 18 potential (h, z) pairs to start with and 324 parent-child permutations.

Forms for transfer paths. In our exercises, the parameters for the insurance transfer function have the following dependencies on time, type, and education status:

$$\alpha_{0,j,t}(\omega) = \alpha_0(\omega) \times \rho_i^t \times \mathbb{I}(t < T - T_c \times \mathbb{I}(j = c)) \tag{A.1}$$

$$\alpha_{1,j,t}(\omega) = \alpha_1 \tag{A.2}$$

$$\alpha_{2,j,t}(\omega) = \alpha_2(\omega). \tag{A.3}$$

In the above, ρ_i controls how the replacement rate declines with age.

We assume the transfer functions for unconditional transfers and transfers given take the following forms

$$\tau_{u,t}(\omega) = u(\omega) \times \rho_u^t \tag{A.4}$$

$$\begin{aligned} \tau_{g,j,t}(\omega) = {} & g_0(\omega) \times \mathbb{I}(t < T + T_c \times \mathbb{I}(j = n)) \\ & + g_1(\omega) \times \rho_g^{(t-T+T_c \times \mathbb{I}(j=c))} \times \mathbb{I}(t \geq T + T_c \times \mathbb{I}(j = n)), \end{aligned} \tag{A.5}$$

with $u(\omega)$, $g_0(\omega)$, and $g_1(\omega)$ being scalars that represent the scale of transfers while ρ_u and ρ_g are parameters that control the degree of front-loading

Parameters for transfer paths. To match the time paths of transfers in the data, we set $\rho_i = 0.975$, $\rho_u = 0.9$, and $\rho_g = 0.95$. We also set the parameters of the transfer functions to the values in table C.1. We choose the values of α_2 to correspond to paying out in the 5% and 25% worst states.

	Insurance			Unconditional		Giving	
Sub-type index	$\alpha_0(\omega)$	$\alpha_1(\omega)$	$\alpha_2(\omega)$	$u(\omega)$	$e(\omega)$	$g_0(\omega)$	$g_1(\omega)$
1	50%	-50	-12.5	\$5,000	50%	\$10,000	\$6,000
2	50%	-50	-5.0	\$10,000	100%	\$30,000	\$15,000
3	100%	-50	-12.5	\$0	0%	\$0	\$0
4	100%	-50	-5.0	–	–	–	–
5	0%	0	0	–	–	–	–

Table C.1: Parameters for transfer functions

C.2 Markov transition matrices with distribution of parent characteristics changing within parent group

In this subsection, we present the Markov transition matrices between parents' and children's wealth and permanent income groups. Whereas in the main text we did not re-rank parents and children after eliminating transfers in either partial or general equilibrium scenarios, these tables present the results with re-ranking.

The experiments in the main text showed the centrality of transfers for children's attainment of particular ranks of wealth or income under the status quo. In other words, they showed that receiving transfers was almost a necessity if one was to be in the top quarter of wealth or income, regardless of who one's parents were. The experiments presented here, in these tables, show how much mobility there is when only endowments of human capital and learning rates are transmitted across generations. They answer the question of how much mobility there is in an economy without transfers.

Tables C.2 and C.2 show the results for retirement wealth and permanent income in the partial equilibrium case. They demonstrate that there is slightly more persistence of ranks in an economy without transfers, though not by much. This tells us that transfers act as a force multiplier for wealth and permanent income. The persistence of ranks and the shape of the distributions are largely determined by the endowments, but transfers strongly magnify the dispersion they generate.

Tables C.4 and C.4 show the results in the general equilibrium case. Ranks are much more persistent in this economy, across almost all wealth groups. Further, except for movement within the top quarter of households and from the bottom quarter into the middle 50%, there is much less upward mobility in this economy. This is intuitive. The sharp rise in wages due to the elimination of transfers means that endowments determine one's destiny to a much larger degree, though individuals at the bottom gain much more from their effectively fixed human capital stock.

		Baseline				No transfers (PE)			
	Child grp.	<25%	25-75%	75-95%	>95%	<25%	25-75%	75-95%	>95%
Parent grp.	<25%	0.51	0.49	0.0	0.0	**0.49**	**0.51**	**0.0**	**0.0**
	25-75%	0.24	0.48	0.21	0.07	**0.2**	**0.57**	**0.18**	**0.05**
	75-95%	0.0	0.5	0.4	0.09	**0.0**	**0.46**	**0.4**	**0.14**
	>95%	0.0	0.64	0.33	0.03	**0.0**	**0.48**	**0.49**	**0.03**

Table C.2: Probability of child retirement wealth group, given retirement wealth group, baseline v. partial eq. counterfactual

		Baseline				No transfers (PE)			
	Child grp.	<25%	25-75%	75-95%	>95%	<25%	25-75%	75-95%	>95%
Parent grp.	<25%	0.53	0.47	0.0	0.0	**0.5**	**0.49**	**0.0**	**0.0**
	25-75%	0.22	0.48	0.22	0.07	**0.19**	**0.59**	**0.17**	**0.06**
	75-95%	0.0	0.5	0.41	0.09	**0.0**	**0.43**	**0.43**	**0.14**
	>95%	0.0	0.66	0.31	0.03	**0.0**	**0.51**	**0.46**	**0.03**

Table C.3: Probability of child permanent income group, given parent's permanent income group, baseline v. partial eq. counterfactual

		Baseline				No transfers (GE)			
	Child grp.	<25%	25-75%	75-95%	>95%	<25%	25-75%	75-95%	>95%
Parent grp.	<25%	0.51	0.49	0.0	0.0	**0.56**	**0.41**	**0.03**	**0.0**
	25-75%	0.24	0.48	0.21	0.07	**0.22**	**0.51**	**0.21**	**0.06**
	75-95%	0.0	0.5	0.4	0.09	**0.01**	**0.5**	**0.41**	**0.08**
	>95%	0.0	0.64	0.33	0.03	**0.01**	**0.58**	**0.37**	**0.04**

Table C.4: Probability of child retirement wealth group, given parent retirement wealth group, baseline v. general eq. counterfactual

		Baseline				No transfers (GE)			
	Child grp.	<25%	25-75%	75-95%	>95%	<25%	25-75%	75-95%	>95%
Parent grp.	<25%	0.53	0.47	0.0	0.0	**0.52**	**0.48**	**0.0**	**0.0**
	25-75%	0.22	0.48	0.22	0.07	**0.21**	**0.55**	**0.2**	**0.03**
	75-95%	0.0	0.5	0.41	0.09	**0.01**	**0.38**	**0.52**	**0.09**
	>95%	0.0	0.66	0.31	0.03	**0.0**	**0.29**	**0.44**	**0.26**

Table C.5: Probability of child permanent income group, given parent permanent income group, baseline v. general eq. counterfactual

C.3 Transfers by wealth group (model v. data)

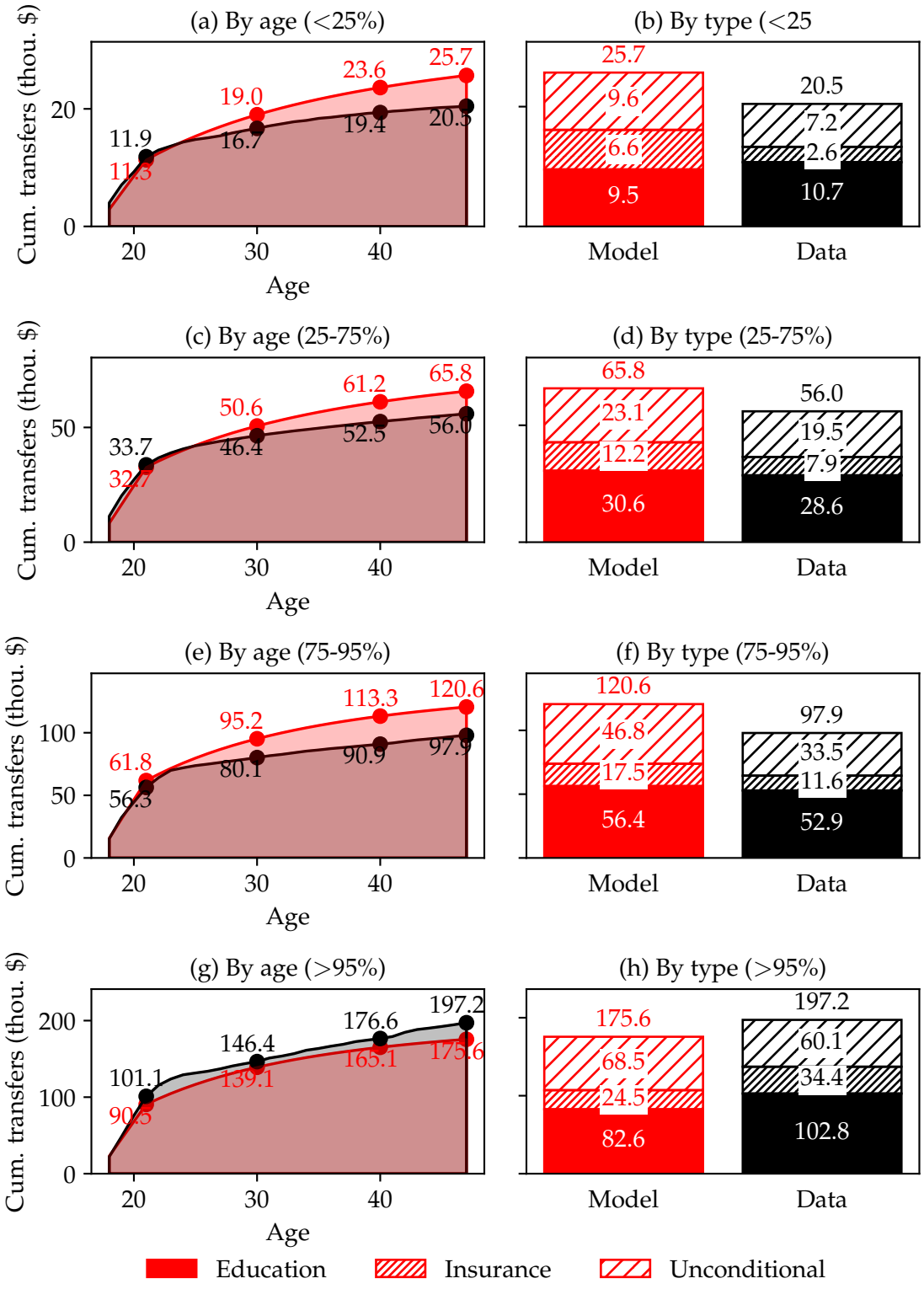


Figure C.1: Transfer paths and transfer mix, by parents' wealth group (model v. data)

D Solving the agents' problems

D.1 Optimal decisions in retirement and the period before

In retirement. In retirement, the path of consumption satisfies the Euler equation

$$c_{t+1} = \left(\beta R_{j,t+1}(\omega)\right)^{\eta_c} c_t \tag{A.6}$$

and the budget constraint

$$a_t + c_t = R_{j,t}(\omega) a_{t-1} + \mathcal{S}(w_j h^\rho) + \tau_{j,t}(\omega). \tag{A.7}$$

With these two equations and the terminal condition $a_{(T_{\ell,j}-1)} = 0$, we may solve for the path of consumption as a function of the starting level of assets $a_{(T_{w,j}-1)}$, the path of income, and the group-specific interest rates. For the purpose of cleaner notation, we'll suppress j subscripts and ωs throughout this derivation.

Solving the budget constraint forward, we obtain

$$c_{(T_\ell-1)} + \sum_{t=T_w}^{T_\ell-2} \left(\prod_{u=t+1}^{T_\ell-1} R_u \right) c_t = y_{(T_\ell-1)} + \sum_{t=T_w}^{T_\ell-2} \left(\prod_{u=t+1}^{T_\ell-1} R_u \right) y_t + \left(\prod_{t=T_w}^{T_\ell-1} R_t \right) a_{(T_w-1)}, \tag{A.8}$$

where we define period income as

$$y_t = \mathcal{S}(w_j h^\rho) + \tau_{j,t}(\omega). \tag{A.9}$$

Substituting the Euler equation (A.6) into (A.8), we find

$$c_{T_w} = \frac{y_{(T_\ell-1)} + \sum_{t=T_w}^{T_\ell-2} \left(\prod_{u=t+1}^{T_\ell-1} R_u \right) y_t + \left(\prod_{t=T_w}^{T_\ell-1} R_t \right) a_{(T_w-1)}}{\sum_{t=T_w+1}^{T_\ell-2} \left(\prod_{u=t+1}^{T_\ell-1} R_u \right) \left(\prod_{v=T_w+1}^{t} d_v \right) + \left(\prod_{u=T_w+1}^{T_\ell-1} R_u \right) + \left(\prod_{u=T_w+1}^{T_\ell-1} d_u \right)}, \tag{A.10}$$

where the object d_t is

$$d_t = \left(\beta R_{j,t}(\omega)\right)^{\eta_c}. \tag{A.11}$$

We may also write (A.10) in terms of present discounted values, so

$$c_{T_w} = \frac{y_{T_w} + R_{T_w} a_{(T_w-1)} + \sum_{t=T_w+1}^{T_\ell-1} \left(\prod_{u=T_w+1}^{t} R_u^{-1} \right) y_t}{1 + \sum_{t=T_w+1}^{T_\ell-1} \left(\prod_{v=T_w+1}^{t} \frac{d_v}{R_v} \right)}. \tag{A.12}$$

Further, the full sequence of consumption levels can be written in terms of c_{T_w} so

$$c_{T_w+u} = \left(\prod_{v=T_w+1}^{T_w+u} d_v \right) c_{T_w} \tag{A.13}$$

and this means that the value starting at the beginning of retirement is linear in c_{T_w}. The full expression is

$$V_{T_w} = \left(1 + \sum_{u=1}^{T_\ell - T_w - 1} \beta^u \left(\prod_{v=T_w+1}^{T_w+u} d_v \right)^{\frac{\eta_c - 1}{\eta_c}} \right)^{\frac{\eta_c}{\eta_c - 1}} c_{T_w}. \tag{A.14}$$

In terms of wealth and income, this is

$$V_{T_w} = \left(1 + \sum_{u=1}^{T_\ell - T_w - 1} \beta^u \left(\prod_{v=T_w+1}^{T_w+u} d_v \right)^{\frac{\eta_c - 1}{\eta_c}} \right)^{\frac{\eta_c}{\eta_c - 1}} \frac{y_{T_w} + R_{T_w} a_{(T_w-1)} + \sum_{t=T_w+1}^{T_\ell - 1} \left(\prod_{u=T_w+1}^{t} R_u^{-1} \right) y_t}{1 + \sum_{t=T_w+1}^{T_\ell - 1} \left(\prod_{v=T_w+1}^{t} \frac{d_v}{R_v} \right)}. \tag{A.15}$$

With the knowledge that $\frac{d_t}{R_t} = \beta d_t^{\frac{\eta_c - 1}{\eta_c}}$, this may be simplified further as

$$V_{T_w} = \left(1 + \sum_{u=1}^{T_\ell - T_w - 1} \beta^u \left(\prod_{v=T_w+1}^{T_w+u} d_v \right)^{\frac{\eta_c - 1}{\eta_c}} \right)^{\frac{1}{\eta_c - 1}} \left[y_{T_w} + R_{T_w} a_{(T_w-1)} + \sum_{t=T_w+1}^{T_\ell - 1} \left(\prod_{u=T_w+1}^{t} R_u^{-1} \right) y_t \right]. \tag{A.16}$$

One period before. Knowing $y_{(T_w-1)} = w_j h^\rho \exp(\epsilon) + \tau_{j,t}(\omega)$ and that, once the value of ϵ is realized, there is no more uncertainty in the model, we may extend equations (A.12),

(A.13), and (A.16) back by one period so

$$c_{(T_w-1)} = \frac{y_{(T_w-1)} + R_{(T_w-1)} a_{(T_w-2)} + \sum_{t=T_w}^{T_\ell-1} \left(\prod_{u=T_w}^{t} R_u^{-1}\right) y_t}{1 + \sum_{t=T_w}^{T_\ell-1} \left(\prod_{v=T_w}^{t} \frac{d_v}{R_v}\right)}. \tag{A.17}$$

$$c_{(T_w+u-1)} = \left(\prod_{v=T_w}^{T_w+u-1} d_v\right) c_{(T_w-1)} \tag{A.18}$$

$$V_{(T_w-1)} = \left(1 + \sum_{u=1}^{T_\ell-T_w} \beta^u \left(\prod_{v=T_w}^{T_w+u-1} d_v\right)^{\frac{\eta_c-1}{\eta_c}}\right)^{\frac{1}{\eta_c-1}} \left[y_{(T_w-1)} + R_{(T_w-1)} a_{(T_w-2)} + \sum_{t=T_w}^{T_\ell-1} \left(\prod_{u=T_w}^{t} R_u^{-1}\right) y_t\right]. \tag{A.19}$$

D.2 Optimal decisions in one's work life

In this section, we provide details for the computations undertaken in order to solve the agent's problem.

Grids and approximations. In solving the model, we will approximate value functions over (a, h, ϵ) for each (h_-, z, ω, j) at each t using the Smolyak method, where the basis functions of the Smolyak polynomial are products of Chebyshev polynomials. We construct the grid and the polynomial using the disjoint-set approach of Judd et al. (2014).

Given our dimensionality $d = 3$ and an approximation level μ, the nested set approach yields a sparse grid over $[-1, 1]^3$, which we will transform to the space $[0, a_{\max}(h_-, z)] \times [h_{\min}(h_-, z, j, t), h_{\max}(h_-, z, j, t)] \times [\epsilon_{\min}, \epsilon_{\max}]$. First, however, we will define the bounds.

Here, $a_{\max}(h_-, z)$ is a maximum level of assets specific to the initial human capital endowment and learning rate (h_-, z), while $h_{\min}$ and $h_{\max}$ are the same quantities, but which depend on endowments as well as education level and years in the workforce. These are defined such that they are the absolute maximum and minimum levels of human capital that one can possibly have in the model. For $j = n$, the starting bounds are

$$[h_{\min}(h_-, z, n, 0), h_{\max}(h_-, z, n, 0)] = [h_- - \Delta, h_- + \Delta], \tag{A.20}$$

where Δ is some appropriately small number. Then, the subsequent bounds can be defined recursively as

$$h_{\min}(h_-, z, n, t) = (1 - \phi)\, h_{\min}(h_-, z, n, t-1) \tag{A.21}$$

$$h_{\max}(h_-, z, n, t) = z\, h_{\max}(h_-, z, n, t-1)^\gamma + (1 - \phi)\, h_{\max}(h_-, z, n, t-1) \tag{A.22}$$

for all the periods $t < T_{w,j}$ and, for $t \geq T_{w,j}$,

$$h_{\min}(h_-, z, n, t) = h_{\min}(h_-, z, n, t-1) \tag{A.23}$$

$$h_{\max}(h_-, z, n, t) = h_{\max}(h_-, z, n, t-1). \tag{A.24}$$

Now, we cover how we transform the Smolyak grid points to this grid. For simplicity, we will suppress any other arguments or subscripts for the bounds. Let $(x_0, x_1, x_2) \in [-1, 1]^3$ be a point on the un-transformed grid. We map x_0 onto the asset space using a single-exponential transformation

$$a = \exp\left(\frac{1}{2}\log(1 + a_{\max})(x_0 + 1)\right) - 1, \tag{A.25}$$

which we adopt for the sake of placing more points around the constraint, which is necessary for the first years after entering the workforce. We map x_1 and x_2 onto the human capital and idiosyncratic productivity spaces using linear transformations, so

$$h = \frac{1}{2}(h_{\max} - h_{\min})(x_1 + 1) + h_{\min} \tag{A.26}$$

$$\epsilon = \frac{1}{2}(\epsilon_{\max} - \epsilon_{\min})(x_2 + 1) + \epsilon_{\min}. \tag{A.27}$$

We may also transform back to $[-1, 1]$, using

$$x_0 = 2\left(\frac{\log(1 + a)}{\log(1 + a_{\max})}\right) - 1 \tag{A.28}$$

$$x_1 = 2\left(\frac{h - h_{\min}}{h_{\max} - h_{\min}}\right) - 1 \tag{A.29}$$

$$x_2 = 2\left(\frac{\epsilon - \epsilon_{\min}}{\epsilon_{\max} - \epsilon_{\min}}\right) - 1. \tag{A.30}$$

The period before the penultimate work period. In this period $t = T_{w,j} - 2$, we solve (12) numerically for each point on the (a, h, ϵ) grid, where $V_{(T_{w,j}-1)}(a', h', \epsilon'; z, \omega)$ is given exactly by (A.19) and we approximate the expectation

$$\mathbb{E}_{\epsilon'}\left[V_{(T_{w,j}-1)}(a', h', \epsilon'; z, \omega)^{1-\eta_r}\right]^{\frac{1}{1-\eta_r}} \tag{A.31}$$

using Gauss-Hermite quadrature, since ϵ' is Normally distributed, with five nodes.

If, instead of optimizing over next period's asset level a' and the current time allocation i, we optimized over the saving rate s and time allocation i, then we are faced with a prob-

lem that involves finding an optimal value, subject to rectangular constraints. We therefore use the multidimensional extension of golden-sections search developed by Sandhya Rani et al. (2019), which we implement in Python.

The output of this step includes an optimal value $\widehat{V}$, a saving rate $\widehat{s}$, and time allocation $\widehat{i}$ for each grid point. The saving rate and time allocations are stored for use in the simulation step, while the optimal values are used to construct the approximation for solving the problem in the period preceding this one.

The periods before that. Solving the next period's problem has produced a value for each point on the Smolyak grid for which we computed an optimum. We approximate the next period's value function using Lagrange interpolation, detailed in Judd et al. (2014).

Our approximating function is a linear combination of the Smolyak basis functions. To find the coefficients defining this combinations, we project the values we found onto the basis functions evaluated at each grid point. Since the basis functions are defined only on $[-1, 1]$, (a', h', ϵ') must be transformed to this space in each evaluation using (A.28), (A.29), and (A.30).

The composition of these transformations with the approximating function defined over $[-1, 1]$ yields the approximation of the next period's value function $V_{j,t+1}$. This function in hand, the process of finding optimal values and policies at each point in this period follows the procedure we outlined to solve the problem in the second-to-last period before retirement.

D.3 Optimal college attendance decision

The functions $\{W_j(h, z; \omega_j)\}_{j \in \{c,n\}}$ are given by equations (9) and (10). By standard results regarding GEV1 shocks, the probability that an agent goes to college, given their parameters, is

$$P(j = c \mid h, z, \Omega) = \left(1 + \exp\left(\frac{W_n(h, z; \omega_n) - W_c(h, z; \omega_c)}{\sigma}\right)\right)^{-1} \tag{A.32}$$

and the expected value of having those particular parameters is given by

$$W(h, z, \Omega) = \sigma \log\left(\exp\left(\frac{W_c(h, z; \omega_c)}{\sigma}\right) + \exp\left(\frac{W_n(h, z; \omega_n)}{\sigma}\right)\right), \tag{A.33}$$

where, in both cases, $\Omega = \{\omega_c, \omega_n\}$.

E Estimating the stochastic processes

E.1 Simulating a life, given initial conditions

Given a starting level of human capital $h_0 = h$, learning rate z, transfer type ω, and education choice j, we can approximate the individual's policy functions for savings rates $s_{j,t}(a_{t-1}, h_{t-1}, \epsilon_t; z, \omega)$ and time allocation $i_{j,t}(a_{t-1}, h_{t-1}, \epsilon_t; z, \omega)$. In the periods strictly before the period before retirement, these will be represented by Chebychev polynomials. In the same way as we did for the value function, we approximate the savings and time-allocation functions by using Lagrange data.

Solving the agent's problem at each (h, z, ω, j) combination yields a saving rate and time allocation for each point of the Smolyak grid in each period. We project these, for a given (h, z, ω, j) and period t, onto the Smolyak basis functions. Then, given a_{t-1}, h_{t-1}, and ϵ_t, we obtain $\widehat{s}_t$ and $\widehat{i}_t$ by evaluating the approximating functions at the current state. Because these interpolated values, $\widehat{s}_t$ and $\widehat{i}_t$, may be be outside the bounds (e.g. negative or greater than one) or violate the budget constraint, we first impose the constraints before using them to produce a_t and h_t. These constraints were also enforced in solving the problem, we simply make sure that they are followed by the interpolated values.[A-1]

For $\widehat{s}_t$, we update according to

$$\widehat{s}_t \leftarrow \min\{\max\left\{\widehat{s}_t, s_{\min,t}(h, z, \omega, j)\right\}, s_{\max,t}(a_{t-1}, h_{t-1}, \epsilon_t; z, \omega), 1\}, \tag{A.34}$$

where s_{min} is the lowest savings rate on the grid (i.e. we don't let the function produce lower savings rates than we ever observe for people with the same qualities in a given period) and s_{max} is the savings rate out of resources needed to put next period's assets above a_{max}, the upper bound for assets.

For $\widehat{i}_t$, we update according to

$$\widehat{s}_t \leftarrow \min\{\max\{\widehat{i}_t, 0\}, i_{\max,t}(a_{t-1}, h_{t-1}, \epsilon_t; z, \omega), 1\}, \tag{A.35}$$

where i_{max} is the maximum number of hours one can work and still keep total earnings positive (i.e. there has to be something left over to consume or save).

[A-1] The interpolation very, very rarely produces values that violate the constraints, though it's wise to allow for them for the sake of robustness in our programs.

After updating, we can compute the next period's state variables using

$$h_t = \delta^t z (h_{t-1} \widehat{i}_t)^\gamma + (1-\phi) h_{t-1} \tag{A.36}$$

$$a_t = \widehat{s}_t (R a_{t-1} + (1 - \mathcal{T}_\ell)\, w h_{t-1} \exp(\epsilon_t)(1 - \widehat{i}_t) + w h_{t-1} f_{j,t}(\epsilon_t, \omega) + \tau_{j,t}(\omega)). \tag{A.37}$$

Starting from $h_{-1} = h$ and $a_{-1} = 0$, we perform these steps until we reach the period before retirement. The steps for simulation become much more straightforward. We can compute sequences of asset holdings and consumption immediately using the formulas of section D.1. For each, (z, h, ω, j) we simulate 10,000 paths, which we use to compute the moments we need.

E.2 Computing conditional expectations

Let q_{65} be the parent's wealth quantile group at age 65. The moments we use to estimate the stochastic processes for human capital, learning ability, and transfer type require us to take expectations of the form $\mathbb{E}[\, y_t \mid q_{65}]$, where y_t is the value of the child's variable y at age t.

Simulation is not effective for computing these expectations inside of a minimum distance problem. It's noisy and inefficient. On the other hand, simulation is effective for computing the expectations $\mathbb{E}[\, y_t \mid h, z, \omega, j\,]$. For each $(h, z, \omega, j\,)$ combination, we can compute this expectation by simulating a large number of individuals with those initial conditions and taking the mean value for that panel. By that principle, we can create N bins for the wealth levels a person can have at age 65 and use simulation to quickly compute $P(g_{a,65} = k \mid h, z, \omega, j\,)$, the probability that a person's wealth bin at age 65 $g_{a,65}$ is the k-th one.

The next step is to marginalize over one's education level j and value for ω. We evaluate

$$P(\omega_c, \omega_n, j \mid h, z) = \underbrace{P(\, j \mid \omega_c, \omega_n, h, z)}_{\text{Output of model solution}} \underbrace{Q_c(\omega_c \mid h, z)\, Q_n(\omega_n \mid h, z)}_{\text{Input processes}} \tag{A.38}$$

$$\mathbb{E}[\, y_t \mid h, z\,] = \sum_{\omega_c, \omega_n, j} \underbrace{\mathbb{E}[\, y_t \mid h, z, \omega, j\,]}_{\text{Pre-computed via simulation}} \quad P(\omega_c, \omega_n, j \mid h, z), \tag{A.39}$$

which allows us compute $P(g_{a,65} = k \mid h, z)$ as well.

The Markov process $Q_{h,z}(\, h, z \mid h_p, z_p)$, which is an input we specify, implies a stationary distribution $P(h, z)$, so we can compute the unconditional distribution $P(g_{a,65} = k)$

along with the unconditional expectation $\mathbb{E}[\, y_t \,]$. Then

$$k_q^* = \min k_q \tag{A.40}$$

$$k_q = \left\{ \ell \,\middle|\, \sum_{k=0}^{\ell} P(g_{a,65} = k) \geq q, \ell \in \{0, \ldots, N-1\} \right\} \tag{A.41}$$

Then we know that $P(q_{65}) = P(g_{a,65} = k_q^*)$ and also that, for the conditional probability, $P(q_{65} \mid h, z) = P(g_{a,65} = k_q^* \mid h, z)$. Moreover, by Bayes' rule

$$P(\, h, z \mid q_{65}) = \frac{P(q_{65} \mid h, z)\, P(h, z)}{P(q_{65})}, \tag{A.42}$$

so we conclude that

$$\mathbb{E}[\, y_t \mid q_{65} \,] = \sum_{h,z,h_p,z_p} \mathbb{E}[\, y_t \mid h, z \,] \underbrace{Q_{h,z}(\, h, z \mid h_p, z_p)}_{\text{Input process}} P(\, h_p, z_p \mid q_{65}). \tag{A.43}$$

This particular procedure shows the case where the parents' wealth group is the top $(1-q)\%$, but we can see how it may be extended quite easily to the other wealth groups we consider in the paper.

E.3 Evaluating the loss and finding an optimum

Taking the model parameters in table 3 as given, the expectations $\mathbb{E}[\, y_t \mid h, z, \omega, j \,]$ and $P(\, j \mid \omega_c, \omega_n, h, z)$ can be pre-computed without any knowledge of $Q_{h,z}$, Q_c, or Q_n. Then, for each evaluation of the loss function, we need only compute the conditional expectations for the relevant moments using the procedure above, which speeds estimation up considerably. It also eliminates simulation error in $(h, z, \omega_n, \omega_c, j)$, since we're able to use the exact distributions over these variables, which massively reduces the difficulty in finding the optimum in this very high-dimensional problem.

Let $m(x, k, t)$ be an entry of an array of moments m indexed by a particular moment x, parental wealth group k, and age t, so that $m(x, t, k) = \mathbb{E}[\, x_t \mid k \,]$. There is an array for the model and an array for the data, though the model's is a function of $\mathbf{Q} = \{Q_{h,z}, Q_c, Q_n\}$. We aim to minimize the RMSE in these quantities and zero-out the total net transfers in

the economy $\mathcal{NT}(\mathbf{Q})$. The total squared loss can then be written as

$$\mathcal{L}(\mathbf{Q}, \mathbf{\Lambda}) = \sum_{x \in \mathcal{X}} \sum_{k \in \mathcal{K}} \Lambda_{x,k} \sum_{t \in T_{x,k}} (m_{Data}(x,t,k) - m_{Model}(x,t,k;\mathbf{Q}))^2 + \Lambda_{\mathcal{NT}} \times \mathcal{NT}(\mathbf{Q})^2, \tag{A.44}$$

where the set of moments is $\mathcal{X} = \{$5y. moving avg. income, cum. educ. transfers, cum. ins. transfers, cum unc. transfers$\}$, the set of parental groups is $\mathcal{K} = \{$<25%, 25-75%, 75-95%, >95%, All$\}$, and the Λs represent weights on each term.

Ultimately, we set each $\Lambda_{x,k}$ to 1, representing equal weights. We also set $\Lambda_{\mathcal{NT}}$ to a suitably large number to indicate we require zero net transfers in the economy. However, in the process of finding an optimum for this loss function, we may adjust $\mathbf{\Lambda}$ to aid the optimizer in moving into more promising areas of the parameter space. In particular, we first put no weight on fitting the paths of transfers and the zero net transfer condition, then gradually increase these over the process of minimization until we eventually hit equal weighting for the moments and very low errors for the zero net transfer condition.